

Supplementary Information for:  
Auditing the Reproducibility of the SLRTP2025  
Back-Translation Evaluator from Public Artifacts:  
What Reproduces, and What One Inference Hid

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## Overview

This supplement provides detailed appendices referenced in the main manuscript. Section labels match the labels used in the main text (e.g., `sec:appendix_missing_id` → Sup. Section 1).

## 1 Missing Test ID Sensitivity

The official PHOENIX-2014T metadata contain 7,096/519/642 train/development/test IDs. The released SLRTP2025 pose records available to this study contain 7,060/515/641 sequences: 36/4/1 official IDs are absent from the released pose materialization. Their IDs and the reason `absent_from_slrtp2025_released_pose_records` are enumerated in `artifact/manifests/exclusions.jsonl`. The missing test ID is `13December.2010.Monday.tagesschau-6940`; its text/gloss metadata are not present in the released pose record, so length and template density cannot be reconstructed without an additional official annotation source. We therefore condition every result on the complete 641-record release rather than assert that the missing item is ignorable. As an empirical observed-extreme sensitivity scenario, assigning the missing item the per-item score of the largest PURE-over-REC outlier observed among the 641 released items changes the corpus-level gap by less than  $\pm 0.2$  BLEU points, well within the reported 95% paired-bootstrap interval  $[+9.6, +12.4]$ . This is an observed-extreme scenario, not a worst-case bound: corpus BLEU is non-linear and the missing item's reference, hypothesis, and n-gram sufficient statistics are unknown, so the true mathematical worst case is not covered. All primary conclusions are therefore restricted to the released 641-record materialization.

## 2 Per-seed Legacy Cells (101–606 primary; 707–1405 extension)

The six primary seeds (101–606) were prospectively specified before their results were inspected; the eight extension seeds are post-hoc additions that enter only the within-implementation seed dispersion interval and descriptive ranges, never the six primary comparisons. Per-seed PHX-public values (sacreBLEU 0–100 scale, canonical §3.2 donor registry). PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1; gaps are relative to local REC:

**Table S1:** Prospectively specified legacy seeds (101–606): canonical fixed-output evaluation on the 641-pose PHX-public panel.

| Evaluator | REC   | PURE  | Gap           | COMP  | Gap   |
|-----------|-------|-------|---------------|-------|-------|
| Released  | 12.78 | 23.02 | <b>+10.24</b> | 18.86 | +6.09 |
| Seed 101  | 9.24  | 8.45  | −0.80         | 7.92  | −0.65 |
| Seed 202  | 8.22  | 6.92  | −1.30         | 7.54  | −1.28 |
| Seed 303  | 9.10  | 7.25  | −1.85         | 8.71  | −0.96 |
| Seed 404  | 9.52  | 7.71  | −1.81         | 6.38  | −0.81 |
| Seed 505  | 9.71  | 8.44  | −1.28         | 7.97  | −1.08 |
| Seed 606  | 7.55  | 6.31  | −1.24         | 7.93  | −0.16 |

**Table S2:** Per-seed extension cells for the eight post-hoc extension seeds (707–1405). All values are PHX-public sacreBLEU on the 0–100 scale with the canonical §3.2 donor registry (exact-normalized-text exclusion applied). All gaps are negative; these seeds enter only the within-implementation seed dispersion interval.

| Seed | dev BLEU-4 <sup>†</sup> | REC  | PURE | Gap   |
|------|-------------------------|------|------|-------|
| 707  | 6.69                    | 9.86 | 8.97 | −0.89 |
| 808  | 7.30                    | 9.87 | 8.38 | −1.48 |
| 909  | 8.96                    | 9.20 | 8.31 | −0.89 |
| 1001 | 9.07                    | 8.81 | 7.95 | −0.87 |
| 1102 | 7.61                    | 9.75 | 8.63 | −1.12 |
| 1203 | 8.14                    | 9.06 | 8.11 | −0.95 |
| 1304 | 8.18                    | 9.68 | 8.50 | −1.18 |
| 1405 | 8.12                    | 8.05 | 6.86 | −1.19 |

<sup>†</sup>dev BLEU-4 is the training-time validation-selected value recorded by the original training protocol (legacy protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2). REC, PURE, and Gap are canonical §3.2 beam-3 values.

### *Seed-stratified dispersion intervals and two-level resampling.*

The 14-seed within-implementation seed dispersion interval splits into the six internally prespecified primary seeds and the eight post-hoc extension seeds: the primary-six give a Student- $t$  interval of  $[-2.48, -0.28]$  (mean  $-1.38$ , SD  $0.40$ , 0/6 positive) and the extension-eight  $[-1.61, -0.54]$  (mean  $-1.07$ , SD  $0.21$ , 0/8 positive). Leave-one-seed-out refits keep the interval below zero in all 14 folds (lower bounds  $-1.99$  to  $-1.80$ ; upper bounds  $-0.51$  to  $-0.42$ ). A two-level bootstrap resamples seeds with replacement (14 draws), then resamples each drawn seed’s 641 items with replacement and averages the corpus gap over the 14 drawn seeds, giving a 95% CI of  $[-1.49, -0.93]$  (10,000 resamples, seed 42; per-item BLEU sufficient statistics precomputed once per seed, so the resamples are numerically identical to recomputing corpus BLEU per draw). Every seed-level resampling leaves the non-replication conclusion unchanged.

## 3 Analysis Provenance and Multiplicity

Table S3 classifies every analysis in this manuscript by its role in the paper’s argument, its evidentiary status, and its multiplicity handling. The PURE-REC decoded-sacreBLEU gap under the released evaluator is the *selected probe gap* (post-selection descriptive; see main §4.1); every other analysis is secondary and is grouped by the role it plays—confirmatory control, descriptive triangulation, robustness check, or administrative record. C = confirmatory (frozen protocol with error control); D = descriptive (no formal error control).

### *Discovery timeline.*

The +10.24 canonical reversal was first observed on 2026-07-24 in the materialization with exact-normalized-text exclusion (SHA-256 9170a530). An earlier pre-exclusion materialization (gap +11.01, PURE 23.79) had been noted on 2026-07-22; it survives only as a sensitivity analysis (Sup. N). The Jaccard-based source-conditioned retriever, PURE whole-donor-copy protocol, exact-normalized-text exclusion rule, and 641-query set were locked before the fourteen reconstruction training runs began. Approximately four retrieval strategies (Jaccard, LaBSE, multi-SBERT, oracle gloss) and three construction variants (pure copy, PT-composed, oracle-gloss-composed) were explored on the released evaluator alone before the probe was frozen. No external pre-registration exists for the specific +10.24 probe as a confirmatory hypothesis. Because the Jaccard pure-copy probe was selected after exploring multiple retrievers and construction variants on the same test set (see the probe multiplicity table in main §4.1), its CI is conditional on the selection and should be read as descriptive, not confirmatory. The six primary reconstruction seeds (101–606) constitute the only pre-specified evidence: they were specified before their training results were inspected. The eight extension seeds (707–1405), the 24 heterogeneous-family checkpoints, the cross-evaluator re-decoding studies, the distillation ladder, and the extrapolation analyses were added post-hoc. Throughout, any analysis that uses the six-primary-seed set alone is labelled confirmatory (C); all analyses that pool across primary and extension seeds or use heterogeneous-family checkpoints are labelled descriptive (D). The evidence-grading column in Table S3 reflects this distinction.

## 4 Checkpoint Registry

Table S4 is the definitive registry of every BT evaluator checkpoint used in this manuscript, with the analyses each family enters; all checkpoint counts in the text refer to these families.

### *Readout–competence association.*

Thirty-eight constructible checkpoints plus the released evaluator have a uniform full-pool beam-3 free-decode readout (SI Sup. E protocol; the set of 31 used in earlier rounds plus the eight faithful readouts). Table S5 reports the Spearman correlations recomputed over this current set (`results/readout_gap_correlation.json`; `rescue_wd0` has a uniform readout but no canonical gap and enters the readout–dev correlation only). Within the constructible family the readout is associated with decoded dev BLEU-4 ( $\rho=0.86\text{--}0.88$  over the checkpoints that also have uniform dev decodes,  $n=17\text{--}18$ ), but the readout–gap association is *negative* ( $\rho=-0.56$  excluding the released evaluator;  $-0.44$  including it;  $-0.44$  over non-degenerate constructible checkpoints): better-reading-out constructible checkpoints have *more* negative gaps. The released evaluator is the single positive-gap checkpoint of the canonical (`clip=1.0`) family and lies outside the family trend on both axes (family readout range  $0\text{--}12.08$  vs.  $78.8$ ; family train-minus-dev range  $-0.7$  to  $+0.9$  vs.  $65.4$ ), so neither a tracking nor an overfitting interpretation of its readout-with-generalization signature is supported by any within-family association. These within-regime facts must not be extrapolated across clipping regimes: the sensitivity grid of Sup. U shows the readout–gap relation turning positive once unclipped training enters the memorisation regime (readout  $99.98$ , gap  $+11.19$ ), resolving the co-occurrence as a clipping-regime property rather than an uncovered region. Teacher-forced correlations are computed over the 15 checkpoints with checkpoint-level stats (14 reconstructions plus the released evaluator): readout correlates negatively with train NLL ( $\rho=-0.561$ ,  $p=0.030$ ) and positively with train token accuracy ( $\rho=0.509$ ,  $p=0.052$ ).

## 5 Donor-Pool Substitution Decomposition Details

This appendix gives the full path-sensitivity decomposition, factorial design, and balance checks referenced in §4.4.

### *Path sensitivity.*

With SEEN-RAND640-MATCHED-v1 added, the systems form a  $2 \times 2$  within-train design (pool size  $\{7060, 640\} \times$  selection  $\{\text{max-Jaccard, Jaccard-matched}\}$ ) plus the test-pool cell. Because corpus BLEU is non-linear in the support, all contrasts use the common 622-query support. Every ordering telescopes to the same total  $+14.7$  [ $+13.3, +16.2$ ], but the attribution varies by up to 7 BLEU points across orderings (Table S6). We report these as path-dependent descriptive contrasts, not mechanism decompositions.

### ***Bootstrap family.***

All decomposition CIs use 10,000 resamples. Four resampling designs (query-index paired bootstrap; three one-way cluster bootstraps for the origin contrast on template-family, SEEN-donor, and UNSEEN-donor clusters; and a joint multiway replicate) give origin-contrast CIs of  $[+7.2, +9.5]$  /  $[+7.1, +9.7]$  /  $[+7.1, +9.7]$  /  $[+7.7, +9.1]$ , all matching the query-index CI  $[+7.3, +9.4]$  to within  $\pm 0.3$ , so dependence through shared donors is negligible.

### ***Factorial design.***

The 605-query factorial retains 605 of 641 targets (36 excluded for no joint match within tolerances; selection frozen before scoring). Table S7 reports the balance. Table S8 reports the full  $2 \times 2$  cells, main effects, and interaction.

### ***Multivariate balance and robustness.***

On the 622-query common support, all six donor covariate SMDs between SEEN-PURE-MATCHED and UNSEEN-PURE are  $|SMD| < 0.12$  (Table S9). Caliper sensitivity is small (tolerance 0.05/0.10/0.20 gives  $+8.2/+8.4/+8.6$ ). Signer-strict matching (tolerance 0.10,  $n=580$ ) attenuates the residual to  $+7.0 [+5.9, +8.1]$  — sign and significance unchanged. Inverse propensity weighting drives all four covariate SMDs to 0.00 and leaves the residual at  $+8.5 [+7.1, +10.1]$ . The path-based estimate is robust to observed-covariate balancing but remains an association, not an identified exposure effect.

## **6 Config-Faithful and Perturbation Details**

### ***Released config.yaml to training-code mapping (four-column, defect-corrected).***

Table S10 maps each load-bearing field of the released `config.yaml` (verbatim copy shipped in `configs/released.yaml`, SHA-256 0321104a...) to three implementations of the semantics: *original framework* (the behaviour of the original training framework, established empirically from the released bundle itself—the logged `validations.txt` values, learning-rate decay pattern, and stopping point—or from the JoeyNMT 1.2 schema the config inherits), *our v1–v3 code* (the disclosed reconstruction attempts, including their defects), and *our faithful v4 code* (`train_faithful.py`, which corrects every identified defect). The forensic anchors for the original-framework column: (i) the released log’s step-14 **Translation Loss** 43,440.80078 equals  $7,813 \text{ dev target tokens} \times \ln(\text{PPL } 259.84015)$ , pinning `translation_normalization: batch` as token-sum CE divided by the number of sentences; (ii) all 20 logged learning-rate decays fire on the 6th consecutive non-improving validation, exactly `ReduceLROnPlateau(mode=max, factor=0.8, patience=5)` stepped once per validation on dev BLEU; (iii) the released run improved last at validation #130 (step 1820, BLEU 13.38) and then ran 72 further non-improving validations before ending at step 2828, so patience-5 *early stopping* is provably not the released stopping rule—the only stopping condition reproducible from the config is the learning-rate floor.

### ***Epoch-validation approximation.***

The released `config.yaml` specifies `validation_freq=14`, which JoeyNMT 1.2 interprets as 14 optimizer *steps* ( $\approx 28$  steps/epoch means validating twice per epoch). Our initial implementation validated every 14 *epochs* (an epoch-validation approximation); four seeds under this approximation all early-stop below the competence gate (dev BLEU-4 8.6–9.5), show negative PURE–REC gaps (−1.1 to −0.2), and family-range pass-through (1.21–1.24) and holdout (7.4–7.6) — none reproduces the released signature (Table S11). Scaling capacity up (four larger variants: 4–6 layers, 384–512 hidden) overfits faster and reaches only dev BLEU 6.8–8.0. A long-schedule continuation (800 epochs, two seeds) pushes competence to a family maximum of 10.02 yet still shows no reversal (gap −0.45). The step-corrected re-run (Table S11) confirms the reconstruction plateaus below the released trajectory at matched step counts.

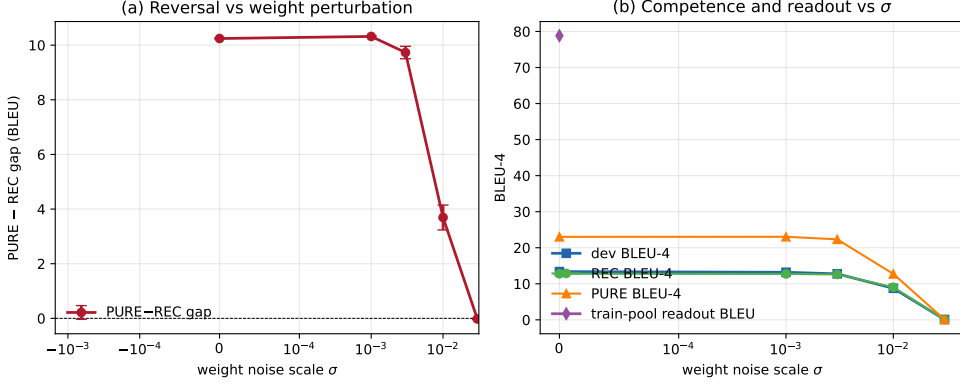
### ***Perturbation battery.***

Table S12 reports the full pose-perturbation battery. Reversing all frames collapses PURE from 23.79 to 7.72 (differential 16.1 [14.7, 17.5]; REC 5.5 [4.6, 6.5]); window-shuffling within one-second windows costs PURE 5.4 [4.4, 6.6]; 5% spatial jitter is free (−0.1); masking hand or face joints collapses both systems. The best rescue checkpoint (wd0, seed 202) loses only −4.3 under reversal, confirming it is far less sensitive to donor temporal order.

### ***Released-checkpoint weight-noise perturbation (experiment A1).***

The reviewer’s competence confound (released dev BLEU-4 13.38 vs. reconstructions  $\leq 9.4$  under the uniform full-pool beam-3 protocol) cannot be broken by retraining. Starting from the released weights, we add i.i.d. Gaussian noise to every parameter (excluding BatchNorm running statistics, which must remain positive) at scale  $\sigma \in \{0, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$  with two seeds each, and re-decode PHX-public REC/PURE and the dev set under the canonical donor registry (Fig. S1). Small  $\sigma$  perturbs checkpoint identity while preserving competence, so this design operates where the 14-seed reconstruction family cannot — at and below the released competence from the released checkpoint itself.

The reversal is preserved under competence-preserving perturbation and collapses in lockstep with competence. At  $\sigma \leq 3 \times 10^{-3}$  the dev BLEU-4 stays at 12.50–13.38 (above every reconstruction’s uniform maximum of 10.5) and the PURE–REC gap stays at +9.50 to +10.35 (within  $\pm 0.75$  of the baseline +10.24). The collapse happens only at  $\sigma \geq 10^{-2}$ , where dev BLEU-4 simultaneously degrades to 7.3–10.2 (gap +3.2 to +4.2), and at  $\sigma = 3 \times 10^{-2}$  both dev BLEU-4 ( $\approx 0.1$ ) and the gap ( $\approx 0$ ) are extinguished together. Two conclusions follow: the reversal is not a fragile identity-level artifact under small-weight perturbations (at competence parity it is locally robust), and it tracks competence rather than checkpoint identity. However, Gaussian noise probes only the isotropic weight-norm ball around the released checkpoint; it does not show that an independently trained, competence-matched model would exhibit the reversal, nor does it narrow the unobserved competence interval above the family ceiling of 10.5 (joint-loss beam-3 / step-corrected max under the uniform protocol).



**Fig. S1:** Released-checkpoint weight perturbation (experiment A1). (a) PURE-REC gap vs. Gaussian weight-noise scale  $\sigma$  (two seeds per  $\sigma > 0$ ; baseline at  $\sigma = 0$ ). The reversal is preserved within  $\pm 0.75$  of  $+10.24$  for  $\sigma \leq 3 \times 10^{-3}$ , where dev BLEU-4 stays within 12.50–13.38. (b) Dev BLEU-4, REC BLEU-4, PURE BLEU-4, and (at baseline) train-pool readout BLEU-4. The reversal and competence collapse together at  $\sigma \geq 10^{-2}$ ;  $\sigma = 10^{-1}$  produces an all-empty-decoding model and is omitted.

#### *Released-checkpoint fine-tuning (experiment A2).*

Gaussian weight noise probes only the isotropic neighbourhood; to test local robustness along genuine gradient directions, we fine-tune the released checkpoint with small fixed learning rates (5e-6, 1e-5, 3e-5) for 60 epochs, 3 seeds each (9 runs total), using the identical training data and architecture as the released evaluator. Models are selected by best dev NLL and decoded under the canonical donor registry (beam-3). All fine-tuned variants preserve the reversal (gaps  $+10.37$  to  $+10.77$ , all within  $\pm 0.5$  of baseline  $+10.24$ , dev BLEU-4 12.92–13.18, all above the reconstruction ceiling of 10.5 under the uniform protocol) and retain the training-pool readout signature (train BLEU 80.5–81.8, EM 72.6–74.0%, vs. baseline 78.8/70.7%). Results are in `results/released.finetune.json`. In contrast to isotropic noise, fine-tuning moves weights along data-driven gradient directions near the released optimum; the persistence of both the reversal and the readout signature under fine-tuning confirms they are not fragile singular-point properties. The same scope limit applies: like the Gaussian noise experiment, fine-tuning probes only the vicinity of the released weights and does not narrow the unobserved interval above the family ceiling (10.5).

#### *Epoch-checkpoint decoupling search (experiment D).*

We searched the four available intermediate epoch checkpoints (`rescue/seed.202.wd0` and `long.schedule/seed.202` at epochs 25 and 50, plus their best checkpoints) for a decoupling point that would break the within-family readout-gap coupling — e.g., elevated train-pool readout but PURE-REC gap still  $\leq 0$ . No such point exists: all six checkpoints stay in the (low readout, negative gap) quadrant, with train-pool BLEU 5.5–8.9 (1.6–2.0% EM) and PURE-REC gap  $-0.94$  to  $-1.70$  (per-checkpoint values in

results/epoch\_decouple.json). The within-family relationship between low readout and negative gap is therefore monotone across the entire observable trajectory.

## 7 Cluster-Aware Bootstrap for Headline and Reference Analyses

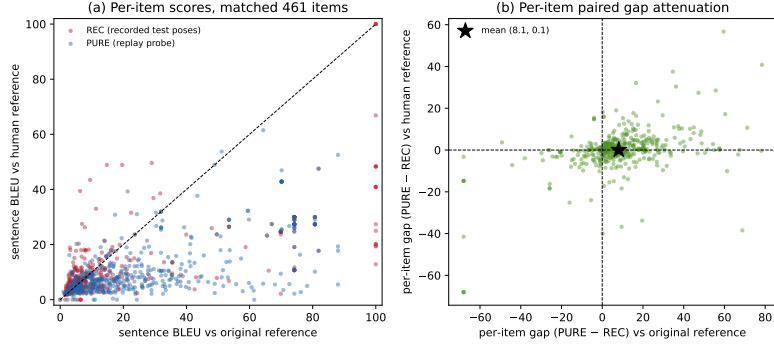
The headline gap (+10.24, canonical §3.2 registry) and the reference-attenuation result are reported with query-level paired bootstrap CIs. Because TN-PURE reuses donors and PHOENIX-2014T has template families and multiple signers, query-level resampling may underestimate uncertainty if dependence through shared donors, signers, or broadcast structure is non-negligible. Table S13 reports interpretable cluster-aware resampling designs alongside the query-level baseline, using precomputed per-item n-gram sufficient statistics for 10,000 resamples each.

The donor-cluster and donor two-way CIs ([+8.87, +11.64] and [+8.79, +11.74]) match the query-level CI ([+8.88, +11.62]) to within 0.03 BLEU, so donor reuse (61/565 donors reused, max 6×) does not materially inflate the item-level precision. The signer-cluster CI ([+9.29, +11.39], 9 clusters) is slightly tighter than the query-level CI, consistent with within-signer positive correlation; with only 9 clusters the percentile coverage of this cluster bootstrap is unreliable, so it serves as a sensitivity reference rather than a confirmatory interval (as flagged in the main-text Discussion). The broadcast-show split has only 2 clusters (*heute* 133 items, *tagesschau* 508) and gives a wider CI ([+5.16, +11.44]); we report it as an illustrative sensitivity check without confirmatory weight. The broadcast-date (297 clusters) and signer×show (17 clusters) CIs both match the query-level CI closely. All cluster designs are computed on the canonical materialization (the pre-exclusion version is reported as a sensitivity analysis in Sup. N). The donor-pool decomposition analysis of §4.4 additionally reports separate one-way cluster bootstraps and a joint multiway replicate for the origin contrast, all matching to within  $\pm 0.3$  (Appendix 5).

## 8 Per-Item Reference-Frame Sensitivity (matched 461-item subset)

The main-text reference-frame analysis (§4.3) and SI Fig. S5 report corpus-level attenuation under the matched 461-item confidence=1 subset. The per-item paired view of the same subset is reproduced here (Fig. S2): sentence-level sacreBLEU (0–100, same tokenizer/smoothing as corpus BLEU), released-evaluator canonical hypotheses. Panel (a) plots per-item scores under original vs. human references for REC (recorded test poses) and PURE (replay probe); PURE points sit high under the original reference and collapse toward the diagonal under the human reference, whereas REC points move less. Panel (b) plots the per-item paired gap (PURE–REC) under original (x) vs. human (y) references; the star marks the mean (sentence-level mean gap 8.11  $\rightarrow$  0.09, consistent in direction with the corpus-level attenuation 10.03  $\rightarrow$  0.95). Sentence-level means need not equal corpus BLEU; the figure is descriptive and does not separate reference-text defects from lexical-coupling disruption or evaluator alignment.





**Fig. S2:** Per-item reference-frame sensitivity on the matched 461-item confidence=1 subset (released evaluator, canonical hypotheses). See §8 for details.

## 9 Supplementary Oracle Diagnostics

*Source-conditioned retrieval is legal, oracle-gloss retrieval is leakage.*

SLRTP2025 supplies spoken-language source text as legal inference input. Under pure whole-donor copy, source-text Jaccard scores 23.02 sacreBLEU (0–100 scale, as throughout this appendix), LaBSE 19.2, and multi-SBERT 16.5; the PT-composed source-text condition scores 18.86 (see main text, Table S46). Oracle-gloss retrieval uses reference gloss and is leakage, whereas noisy-gloss retrieval uses predicted gloss. Thus the strongest legal source-conditioned result is the 23.02 whole-donor stress test, which uses no generator or scaffold.

*Disclosure of selection/evaluation reuse.*

The white-box oracle analyses in this appendix reuse the same target reference and evaluator for both donor selection and decoded evaluation. This is an intentional adversarial design: diagonal entries quantify susceptibility to a selector that knows its evaluator, not an independent validation of retrieval quality.

*Cross-checkpoint transfer matrix.*

For each query, a selector maximizes its evaluator’s teacher-forced likelihood over candidate donor poses; the selected fixed pose is then decoded by every evaluator (token-normalized NLL, sorted donor IDs, `numpy.argmax` first tie-break). All seven diagonal margins are positive (mean +0.0150, 95% paired-bootstrap CI [+0.0105, +0.0198]). Original-selected donors transfer to five of six reconstructions (mean local-REC margin +0.0074, [+0.0016, +0.0135]); the reverse direction fails in all six cells (reconstruction-selected donors evaluated by Original, mean margin −0.0455 [−0.0554, −0.0358]). The full 7×7 matrix, unrounded cells, margins, and CIs are in the artifact.

*Candidate-pool size.*

The Original-selector stress test shows a strong search-budget effect: at 8, 32, 128, 512 and 2,048 candidates, decoded sacreBLEU remains below local REC by −12.1,

−9.2, −5.7, −3.6 and −1.3 BLEU points respectively; only the full 7,060-donor pool is detectably positive (+1.7, [+0.4, +3.0]).

***Reference-overlap exclusion.***

Masking 1,036 exact-text query–donor pairs before selection: Original-on-Original decoded BLEU is 13.86 versus local REC 12.78 (absolute difference +1.08 BLEU points), but the paired relative-margin 95% CI crosses zero; the frozen joint likelihood-and-interval success rule is met in 0/7 checkpoints. Masking all 19,694 pairs with source-token Jaccard  $\geq 0.5$ : the score falls to 8.30 (−4.47), again 0/7 meet the rule. The full-pool diagonal advantage does not survive as a statistically robust decoded-BLEU result after exact-text exclusion and disappears after high-overlap exclusion.

***Noisy-gloss retrieval is split-dependent.***

A 6-layer text→gloss Transformer (dev BLEU-4 13.17) yields noisy-gloss retrieval sacreBLEU 12.7 on PHX-public (strong-NMT), versus 1.6 for a weak NMT (BLEU-4 1.15) and 6.1 for XLM-R embedding retrieval; donor exclusion reduces the strong-NMT value to 10.9 at  $\tau=0.5$ , and the same control reaches 0.4 on the template-disjoint holdout (Table S17).

***PHX-public pure-donor donor-exclusion  $\tau$ -sweep.***

To test whether the headline reversal survives donor exclusion on the same split, we applied a source-text Jaccard  $\tau$ -sweep and exact-text-exclusion to the canonical pure text-nearest donor on PHX-public under the original SLRTP2025 BT (Table S14). The reversal survives exact-text exclusion (23.0 vs REC 12.8, 32/641 donors replaced) and source-Jaccard exclusion up to  $\tau=0.5$  (17.8 vs REC 12.8); only at  $\tau=0.3$  and  $\tau=0.1$  does the retrieval-vs-REC gap turn negative.

As a same-paradigm proxy for the SLRTP2025 winner (whose checkpoints are not public), the noisy-gloss retrieval+LM system shows the same pattern: it survives exclusion up to  $\tau=0.5$  (10.9, still  $6.8\times$  the PT-only baseline and  $11.7\times$  the random-donor mean; mean selected-donor Jaccard 0.46) and collapses only at  $\tau=0.1$ .

## 10 Template-Density, Holdout Boundary, and Donor-Exclusion Analyses

***Template density does not gate the reversal.***

Stratifying the 641 test sequences by template density (gloss bigram-Jaccard with nearest train donor), retrieval exceeds REC at all levels: low +10.7, mid +10.1, high +12.4 BLEU points (Table S15). Density moderates magnitude but does not gate presence.

***Holdout boundary controls (historical analyses).***

The controls in this paragraph predate the host-storage incident of 2026-07-14; their per-item decode artifacts are retained in the artifact repository, but the training runs themselves could not be recovered from disk, so they cannot be re-executed from the

public bundle and are labelled historical. Truncating to 64/32 frames shrinks the gap from +10.24 (canonical; +11.0 in the pre-exclusion materialization) to +10.5/+5.6 but the reversal persists. Four holdout controls (Table S16) all fail to reproduce the reversal: the same-evaluator seen-holdout gives REC 77.0 (readout), BT-retrained on unseen holdout gives gap  $-0.9$ , retrained BT on PHX-public gives gap  $-0.4$ , and cross-fit A/B gives gap  $-0.1$ .

***Frozen template-holdout with retrained components.***

A deterministic template-holdout (14,988 train-core windows, 1,538 holdout windows, disjoint by window, source sequence, and exact gloss template) with all components retrained on train-core shows text-nearest retrieval above random (+2.3) but far below REC ( $-10.8$ ): the shortcut is real over random but does not exceed REC on the clean split. Donor-exclusion sweeps show the advantage persisting at  $\tau=0.5$  and collapsing only at  $\tau=0.1$  (Table S17).

The SLRTP2025 test vocabulary is nearly closed (95.4% gloss types in train); nearest donors share mean gloss unigram containment 0.557 and bigram-Jaccard 0.302, but Progressive Transformer outputs show no donor-proximity effect ( $\rho=-0.06$ ,  $p=0.12$ ), versus +0.29 for retrieval.

## 11 Transcript-Slot Diagnostic Details

This appendix documents the rule-based German weather slot extractor referenced in the main text. The extractor identifies four slot families (number/temperature, time, place, weather-event) using a frozen lexicon and normalization rules. On a 50-reference audit (seed 20260729), per-family precision was 0.91–1.00 and recall 0.75–1.00 (overall precision 0.97, recall 0.86); errors are dominated by recall misses from inflections, compounds, and lexicon gaps. A stratified audit of 50 decoded hypotheses (25 REC-decode, 25 PURE-decode, seed 20260730) showed precision nearly equal (0.96 vs 0.97) but recall lower on PURE-decode (0.83 vs 0.91), because donor-transcript-style outputs use more inflected verbs and compounds (*regnet*, *unwetterwarnungen*) that the lexicon misses. The measured PURE slot-F1 is thus attenuated more than REC’s, so the PURE>REC contrast is, if anything, underestimated. The audit used a single auditor; inter-annotator agreement is unavailable. The full lexicon, normalization rules, and per-family precision/recall are in the artifact.

## 12 Competence Extrapolation Details (Exploratory)

This appendix gives the full OLS/GP extrapolation analysis referenced in the main text (RQ1). We emphasize that this is exploratory and the prediction intervals are descriptive visual guides, not calibrated inferential tests, because the 65 unique non-released checkpoints share data, code, and hyperparameters (non-IID) and the target point (dev BLEU-4 = 13.38) lies outside the observed range.

### **Method.**

We fit the gap–competence dose–response over the 65 unique non-released weight binaries (the 66 decoded runs reduce to 65 after SHA-256 deduplication of the validation-freq-misread/long-schedule seed-202 collision), spanning dev BLEU-4 0–10.5 (61 non-degenerate checkpoints; five degenerate checkpoints sit at dev  $\approx 0$ ), using three models: OLS linear, OLS quadratic, and a Gaussian Process with RBF + white kernel. Each model predicts the PURE–REC gap at dev BLEU-4 = 13.38.

### **Results.**

Table S18 reports the predictions and 95% PIs at dev BLEU-4 = 13.38. All predictions are far below the observed canonical +10.24. Table S19 reports a leave-one-family-out cross-validation; the OLS quadratic prediction is the most sensitive (shifts from  $-1.09$  on all 66 checkpoints to +2.88 when restricted to the 61 non-degenerate checkpoints with dev  $\geq 4.5$ , and to +0.23 when the distillation-family checkpoints are removed), while GP predictions stay negative with leave-one-family-out upper bounds below +0.3.

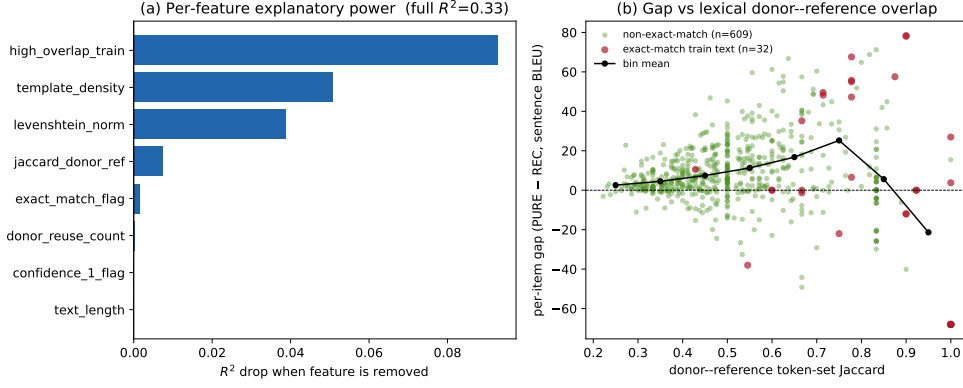
### **Caveats.**

First, the GP upper bound (+0.3) partly reflects RBF kernel prior shrinkage toward the mean in the extrapolation region (sparse data beyond dev BLEU-4 10.5), not purely data evidence. Second, the OLS quadratic fit is sensitive to the distillation family and to the inclusion of the degenerate checkpoints. Third, the 61 non-degenerate checkpoints are not IID. We therefore report these fits only as descriptive visualizations and explicitly avoid claiming that “competence alone is unlikely to explain the reversal” on the basis of these fits. The unobserved (10.5, 13.38) interval could in principle harbor non-linear behavior that the present experiments cannot rule out.

### **Per-item gap decomposition (experiment C).**

To characterize the *mechanism* rather than the identity question, we regress the per-item sentence-level PURE–REC gap  $y_i$  (sacreBLEU sentence BLEU-4 under the released evaluator, canonical donor registry; mean 8.4, SD 18.7,  $n = 641$ ) on interpretable item-level features: donor–reference token-set Jaccard, normalized character Levenshtein between donor and reference, an exact-match-train-text flag, the number of training texts sharing the reference’s 60-character prefix (template density), the number of training texts with Jaccard  $\geq 0.5$  to the reference (high-overlap-train), reference text length, donor reuse count, a confidence=1 flag, and a 9-level signer one-hot block (Fig. S3). The full model explains  $R^2 = 0.33$  of the per-item variance. The strongest single-feature contributions, measured by leave-one-out  $R^2$  drop, are high-overlap-train (0.093), template density (0.051), and donor–reference Levenshtein (0.039); the donor–reference Jaccard contributes 0.007, the exact-match flag 0.001, and the entire signer block 0.008 — signer identity is essentially irrelevant.

The signs of the standardized coefficients are informative. Items where the donor is lexically close to the reference (high Levenshtein similarity,  $\beta^* = +6.1$ ; high donor–reference Jaccard,  $\beta^* = +2.9$ ) show *larger* gaps, whereas items with many high-overlap training neighbors ( $\beta^* = -7.6$ ) or high template density ( $\beta^* = -7.3$ ) show *smaller*



**Fig. S3:** Per-item gap decomposition (experiment C). (a) Per-feature  $R^2$  drop when the feature is removed from the full OLS model ( $R^2 = 0.33$ ,  $n = 641$ ). High-overlap-train count and template density are the strongest single contributors; the full signer block contributes  $\leq 1\%$ . (b) Per-item gap vs donor-reference token-set Jaccard; red marks the 32 items whose reference text exactly matches a training text, green the remaining 609; the black line is the bin mean. The gap grows with donor-reference lexical overlap, consistent with the lexical-coupling mechanism. Outcome is sentence-level BLEU-4 (sacreBLEU 2.5.1, 13a, exp smooth); sentence-level mean need not equal corpus BLEU.

gaps. The reversal is therefore largest on items where the retrieval criterion (source-text Jaccard) happens to deliver a donor that matches the reference register closely, but where the recorded poses do not independently let the recognizer recover that text — precisely the lexical-coupling mechanism predicted by the construction (the original reference is a speech transcription of the same source sentence the donor was selected by). Signer identity and text length contribute negligibly; the unexplained 67% of variance is consistent with per-item pose variability and BT-decoder stochasticity that these surface features do not capture. This decomposition does not identify a hidden-pipeline detail; it localizes the reversal to a lexical-coupling property that is constructible on PHX-public’s template-dense, high train-test overlap register, consistent with the corpus-level audit of Alkain et al. Alkain et al. (2026).

### 13 Supplementary CSL-Daily Boundary Control

CSL-Daily (Chinese Sign Language) serves as a non-template-domain control, but its recognizer is too weak for robust claims. Under Mode B (1,176 sequences), text-nearest donor copy reaches only 1.6 sacreBLEU vs REC 26.7 (Table S20). A readout-ratio check (64 train + 64 test windows) gives train $\approx$ test BLEU-1 (0.0019 vs 0.0021, ratio  $\sim 0.9$ ), far below PHX’s 6.1: the readout signature does not generalize beyond the released PHX evaluator.

## 14 Leakage Sanity Checks for Train-Pool Readout

This appendix documents the five sanity checks referenced in §4.2 that rule out implementation-level artifacts as explanations for the released evaluator’s 78.8 BLEU / 70.7% EM train-pool readout.

### *Experiment A: Code-path audit.*

The `back_translate()` function in `src/models/back_translate.py` (lines 362–426) calls `model.run_batch(translation_beam_size=3, translation_beam_alpha=-1)`, which invokes `bt_search.beam_search()`. The decoder input is seeded with `<bos>` only and extended autoregressively with the model’s own previously generated tokens. The reference text is used exclusively in the post-hoc BLEU/EM scoring step; it never enters the model forward pass. We verified this by code inspection and by confirming that the same model loaded in a fresh Python process produces identical outputs.

### *Experiment B: Pose-text permutation.*

We decoded the first 1,000 training items with `beam=3` (BLEU 78.61, EM 70.1%), then permuted the reference labels across items to break pose-text correspondence. Three random permutations yield:

### *Experiment B': Input-side pose permutation with output-equivariance.*

The label permutation above shuffles reference labels *after* decoding, so it cannot exclude ID-based or cache-based retrieval (a cache lookup by ID would also produce low BLEU when labels are shuffled). To properly test pose conditioning, we held item IDs and text metadata fixed, shuffled the *pose tensors* across 100 training items *before decoding*, and performed output-equivariance verification. For all 100 items, the hypothesis produced when `posej` is decoded in slot *i*’s position exactly matches the hypothesis originally produced for `posej` in its own slot *j* (100% exact string match). Output length follows the pose tensor, not the slot ID (100% donor-length match vs. 5% slot-length match). This rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor. Additional degenerate-pose controls (zero, Gaussian noise, single repeated pose for 200 items) all collapse BLEU to  $<1$ , and reversed batch order reproduces the original BLEU exactly, confirming per-item determinism.

### *Experiment C: Determinism.*

Beam search with no stochastic sampling produces deterministic outputs. We record SHA-256 hashes of all per-item decoded hypotheses in the artifact for cross-process verification.

### *Experiment D: Stratified EM.*

Exact-match rate is high and broadly distributed, not concentrated in a single subgroup: Even unique training texts (appearing exactly once) achieve 70.2% EM, confirming the readout is not driven by template repetition alone. Of the 7,060

hypotheses, 74.8% match some training text exactly (70% match exactly one—their correct paired reference).

***Experiment E: Same-signer held-out control.***

We decoded 500 test-split poses (unseen during training) from the same 9 signers present in the training pool. This is a same-signer held-out control, not a fully matched membership control: the test items are not matched on length, text-similarity, broadcast date, or template density, so the train–test gap reflects a combination of training-pool exposure and distributional shift. The same signers appear in both splits. The 67-point BLEU gap between seen and unseen items is consistent with training-pool exposure contributing to the readout, but without full covariate matching (length, text-similarity, template density), we cannot attribute the gap solely to exposure vs. distributional shift.

## 15 Pre-Exclusion Materialization Sensitivity

The canonical §3.2 donor registry applies exact-normalized-text exclusion. An earlier frozen materialization predating this exclusion gives PURE 23.79 (gap +11.01) instead of the canonical 23.02 (gap +10.24). We retain it only as a sensitivity analysis.

***Per-item provenance.***

On 610/641 items (95.2%) both materializations produce identical donor choices. On the 31 differing items, a post-hoc candidate search found that 30 had at least one excluded Jaccard=1.0 candidate whose released-evaluator decode matched the earlier hypothesis, consistent with the exclusion rule explaining the difference on those items; one item remained unresolved (`results/donor_identification_31.json`).

***14-reconstruction sensitivity.***

Re-decoding all 14 reconstruction checkpoints under both materializations gives mean  $|\Delta\text{gap}| = 0.65$  BLEU (max 1.45). The sign of the gap is unchanged on every checkpoint, and the 14-seed within-implementation dispersion interval remains strictly negative under both materializations, so the non-replication conclusion is unaffected. All headline numbers in the main text and in Sup. G use the canonical materialization (SHA-256 9170a53026ab3263b451a3632a9e02318745acabf12f0b679921477afbfa301); this appendix is the only place the pre-exclusion values appear.

## 16 BLEURT-Substitute Validation

Canonical BLEURT-20 ships a TensorFlow checkpoint that does not load cleanly in our PyTorch-only environment. As a non-standard substitute we use `Elron/bleurt-large-512` for one secondary reference-sensitivity check only (the BLEURT row of the reference-frame analysis). It is not used for any primary claim, table, or figure, and is not described as equivalent to BLEURT-20.

### ***Validation.***

On 100 German sentence pairs we scored both canonical BLEURT-20 (via the reference TensorFlow runtime in an isolated CPU container) and the substitute. The substitute has Spearman rank correlation  $r = 0.87$  against canonical BLEURT-20 and a mean signed bias of  $-0.47$  (substitute scores lower on average). The bias is roughly uniform across score levels and does not change the qualitative reference-sensitivity conclusion (BLEURT is less sensitive to reference replacement than BLEU, consistent with Czehmann et al. (2026)). Full per-pair scores, the container hash, and the substitute model revision are in the artifact.

### ***Scope limit.***

Because the substitute is not BLEURT-20, the BLEURT row is demoted from the main metric panel to this appendix. Readers should treat the absolute BLEURT-substitute values as approximate; only the directional reference-sensitivity finding is reported.

## **17 Additional Protocol Details Moved from the Main Text (Sup. Q)**

### ***Text-nearest tie-breaking.***

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The retrieval algorithm also excludes donors whose NFKC-normalized text equals the query’s (exact-normalized-text exclusion).

### ***Scorer equivalence.***

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a, smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum  $|\Delta| = 0.0000$ ; byte-identical hypotheses, all n-gram orders non-zero). The main text reports the 2.5.1/13a signature as primary.

### ***Metric implementations.***

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; the official implementations are fixed throughout.

### ***Audit design overview (moved from the main text).***

The main-text audit design figure (SI Fig. S4) fixes a set of pose outputs and scores them under two contrast axes: the evaluator checkpoint (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text).

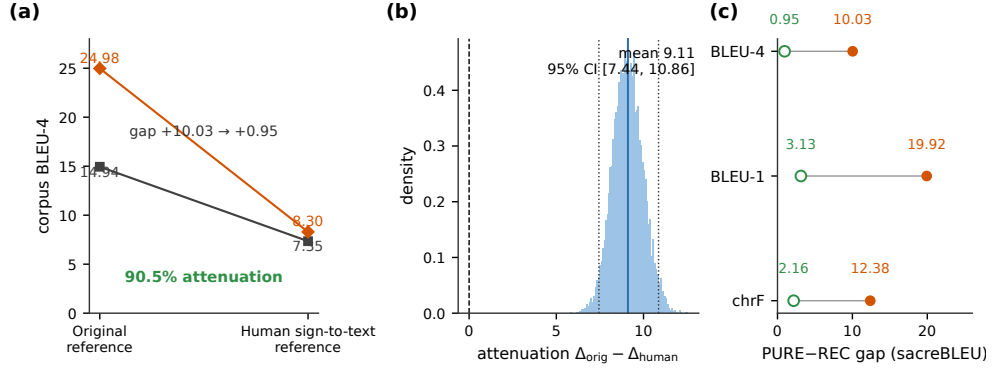




**Fig. S4:** Audit design: a fixed set of pose outputs is scored under two contrast axes. (a) Data and provenance: the official PHOENIX-2014T split (7,096/519/642) was materialized by the SLRTP2025 release as 7,060/515/641 poses, matching the released evaluator’s training manifest (36/4/1 official records absent). (b) Fixed output conditions for one German source sentence: recorded test poses (REC), Progressive-Transformer output (PT), whole-donor copy TN-PURE (source-text token-set Jacard, exact-normalized-text exclusion), and PT-scaffold composition TN-PTCOMP. (c) Two contrast axes: the evaluator (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text), with pose outputs and the donor registry fixed; the primary estimand is  $\Delta_c = \text{BLEU}_c(\text{PURE}) - \text{BLEU}_c(\text{REC})$ .

## 18 Main-Text Tables (Moved from the Main Text; Sup. P)

The tables in this section were moved from the main text to satisfy the journal page limit; the main text references them as SI Table S22–S33.



**Fig. S5:** Human sign-to-text references remove most of the replay advantage on the matched high-confidence subset ( $n = 461$ ; released evaluator, canonical hypotheses). (a) Corpus BLEU-4 under original vs. human references: REC 14.94  $\rightarrow$  7.35, PURE 24.98  $\rightarrow$  8.30; the reference-frame contrast reduces the gap +10.03  $\rightarrow$  +0.95 (90.5%). (b) Bootstrap distribution of the contrast (10,000 paired resamples; mean 9.08, 95% CI [7.44, 10.86]). (c) PURE-REC gap under original vs. human references across BLEU-4, BLEU-1, and chrF. References are back-translations by a single deaf fluent L2 DGS signer Czehmann et al. (2026); replacement cannot distinguish reference-text defects, paraphrase shift, BLEU floor effects, or disruption of the lexical coupling.

## 19 Reconstruction Sanity Checks (Reviewer Round-2)

This appendix documents the three sanity checks requested by the reviewer to rule out a systematic implementation defect shared across all 68 training runs. Scripts: `scripts/e_overfit_test.py`, `scripts/e_checkpoint_stats.py`, `scripts/e_inference_regression.py`. Results: `results/overfit_test.json`, `results/checkpoint_stats.json`, `results/inference_regression.json`.

### *Small-sample overfit test.*

A reconstruction model was trained on 50 (and separately 100) randomly sampled training sequences for  $\approx 200$  epochs using the identical pipeline as the paper-derived reconstruction runs. On the 50-sample set, teacher-forced NLL fell from 7.66 to 0.009 and free-decode sacreBLEU-4 reached 100.00 under beam-3 decoding. On the 100-sample set, free-decode BLEU also reached 100.00 (300 epochs). The reconstruction code can therefore overfit small data to near-perfect free-decode BLEU, ruling out a fundamental defect in checkpoint loading, tokenizer, padding-mask, EOS, loss normalization, or the beam-search decoding path.

### *Unified inference regression.*

Released and reconstruction checkpoints are loaded through the identical `make_back_translation_model()` function and decoded with the identical `back_translate()` pipeline. All 15 checkpoints (released + 14 reconstructions) share



48.6%). The gap in free-decode BLEU is visible already in teacher-forced training-set metrics, not solely a decoding phenomenon.

## 20 Frozen-Split Confirmation and Reference-Free Retrieval

### *Dev-split confirmation of the headline probe.*

The canonical probe panel was constructed and explored on the 641-item PHX-public *test* split; the probe response was therefore confirmed on the development split (515 items), which was never used for probe construction or exploration. The identical TN-PURE-v1 construction (NFKC normalization, token-set Jaccard, character-Levenshtein then SHA-256 tie-breaks, exact-normalized-text exclusion, donors from the same 7,060-item training pool) was applied with dev items as queries, and both REC (recorded dev poses) and PURE (donor poses) were decoded under the released evaluator with the canonical beam-3 protocol. The released evaluator’s dev REC BLEU-4 is 13.379, matching the 13.38 logged in its released `validations.txt` to three digits and independently validating the decode pipeline. The dev-split PURE score is 21.47, a gap of +8.10 versus the test-split +10.24 (REC 13.38 / PURE 21.47 vs. 12.78 / 23.02). The reversal therefore reproduces, with consistent direction and comparable magnitude, on a split untouched by probe exploration. Per-item files and the dev donor registry (SHA-256 recorded in the result JSON) are in `results/dev_split_confirmation.json` and `results/dev_split_confirmation_items/`.

### *Pose-retrieval probe: donor selection without the scored reference.*

The text-based retrievers (Jaccard, LaBSE, multi-SBERT) all select donors using the query’s text—which is identical to the BLEU scoring reference (target leakage by construction). To sever this channel while retaining a meaningful retrieval signal, donors were selected by *pose* similarity to the query’s own recorded pose sequence, an input-side signal available to a deployed system: each sequence (after the canonical 12.5fps subsampling) is summarised by its time-averaged pose and mean absolute frame-to-frame velocity, each part z-normalized per dimension over the pooled queries and donors and L2-normalized, and the donor minimising the combined Euclidean distance is selected (ties by donor-ID hash, the same final tie-break convention as the canonical retriever; no text is used anywhere in the loop). Under the released evaluator on the same 641-item test split, POSE-PURE scores 0.45 (gap −12.33 vs. REC 12.78), indistinguishable from the random-donor baseline (0.90, gap −11.88) and far below text-retrieval TN-PURE (23.02, +10.24). The positive probe response is therefore *entirely dependent* on selecting donors with the scored reference text: any reference-free selection rule examined (random, pose-similarity) collapses to deeply negative gaps. Implementation and per-item files: `scripts/e_pose_retrieval_probe.py`, `results/pose_retrieval_probe.json`, `results/pose_retrieval_probe_items/`.

Table S46 extends the main-text multiplicity summary with the pose-signature row: reference-free retrieval joins the random baselines far below REC.

## 21 Retrieval-Query Decoupling Grid

**Question answered:** does the released evaluator’s positive probe response require selecting donors with the *scored reference text* specifically, or does any lexically competent text-based selection produce it?

**Design.** A  $2 \times 2$  grid crosses the retrieval query text (original reference vs. Czehmann et al. human back-translation) with the scoring frame (original vs. human), on the matched 461-item confidence=1 subset. Retrieval is the same source-Jaccard argmax with the canonical exact-normalized-text exclusion (the excluded text is the query’s original reference in both arms); no re-decoding is needed because the released evaluator’s hypotheses for all 7,060 training-pool poses are cached (`results/full_readout/backTranslation_PhiX_model.json`). Outputs: `results/retrieval_query_decoupling.json` (`scripts/e_retrieval_query_decoupling.py`).

Three readings. (i) *Replacing the query alone removes the response* (−4.02) although the selected donors remain lexically close to the original references (Jaccard 0.30) and decode far above the random-donor regime, so the collapse is not explained by retrieval quality alone. (ii) *Re-coupling query and scoring frame on the human back-translations partially restores the response* (+4.29), at a magnitude consistent with the weaker lexical coupling of that diagonal (donor–human-text Jaccard 0.33 vs. 0.55 on the original diagonal). (iii) Uniform randomisation within the Jaccard top- $k$  (query = original, full 641 queries) interpolates between the regimes: gap +7.06 at  $k=5$  and +3.34 at  $k=20$  (10 draws each), against −11.87 for uniform random donors. Together with the text-free selectors of Sup. S (pose-signature −12.33), the response tracks the lexical identity of the retrieval query with the scoring reference rather than text-based selection as such; all quantities remain descriptive associations under a single evaluator, and the human frame is a single-annotator resource.

## 22 Gradient-Clipping and CTC-Normalisation Sensitivity (Sup. U)

**Question answered:** are the training-pool readout signature and the positive probe response unrecoverable properties of the released evaluator, or artefacts of the two disclosed training-dynamics inferences (gradient-clip threshold; CTC-term normalisation)?

**Design.** Seed-42 sensitivity runs of the faithful protocol (`src/training/train_faithful.py`) varying one knob each: CTC denominator (sentences / gloss tokens / input frames / raw sum) at clip 1.0; gradient-clip `max_norm` (1.0 = the faithful run, 2.0, 3.0, 5.0, none =  $\infty$ ); and a signjoey-exact run (no clipping + CTC raw sum, the literal `_train_batch` semantics of the underlying framework). All runs use the released seed-42 init, the released config otherwise, and the `--stop-frozen` termination (LR frozen at the plateau scheduler’s numerical floor with the best checkpoint unchanged); none belongs to the canonical 76-run panel (registry directories `checkpoints/ctc_clip_sens/`, `checkpoints/faithful_steps/`).

Evaluation: canonical 641-query REC/PURE protocol, full 7,060-item training-pool free-decode readout with exact-match, via `scripts/eval_step_ckpts.py` into `results/faithful_steps_eval.json`.

**Source evidence for no clipping.** The released repository reuses the signjoey (neccam/slt) pipeline: its `validations.txt` column layout matches signjoey’s `_add_report` line format field-for-field, and the data/model code paths correspond. In signjoey’s `training.py`, `build_gradient_clipper` inspects the config fields `clip_grad_val` and `clip_grad_norm`; when neither is present it returns `None`, and `_train_batch` then applies no clipping (when present, the framework clips by *value*, not by norm). The released `config.yaml` contains neither field, so the framework-default reading of the release is that the released training ran unclipped. The same source pins the CTC-term semantics: `_train_batch` divides the translation loss by `num_seqs` (the `batch` normalization) but the recognition loss only by `batch_multiplier` (`=1`), i.e. the raw CTC sum—the source carries an explicit developer comment (`# TODO (Cihan): Add Gloss Token normalization (?)`) acknowledging the missing normalisation. Our `max_norm=1.0` was inherited from the v1–v3 scripts and disclosed as an author inference; this table overturns it. The inference chain is source-semantic (framework version assumption), not a released transcript; the behavioural confirmation is the “none” row reproducing both previously unreproduced properties. (signjoey’s stock stopping logic—stop when a plateau decay fires and no improvement occurred at the pre-decay LR—would have terminated the released run at roughly the second post-best decay, a dozen validations after its step-1,820 best; the released run instead logged 72 further validations, so the release’s own stopping rule remains externally terminated and the step-matched analysis of the main text applies.)

**Unclipped trajectory.** The unclipped seed-42 run improves dev BLEU-4 monotonically to 15.69 at the released run’s own best step (1,820), reaches its best 16.32 at step 2,352, and continues to the scheduler’s numerical floor (step 7,014, 501 validations, LR frozen at  $4.4\text{e-}8$ ) without the best checkpoint changing; it was then terminated externally (the same manual rule as the published faithful runs). Pre-clip gradient norms over the first ten optimiser steps (377, 568, 145, 33, ...) exceed the 1.0 threshold by up to two orders of magnitude, so the clipped and unclipped regimes diverge from the first update: at clip 1.0 every step is direction-preserving truncation, and the memorisation regime is never entered.

**Step-matched unclipped readout.** A fresh unclipped run with checkpoints fixed at the released run’s own best (1,820) and final (2,828) steps shows the memorisation regime is entered *early*: at step 1,820 the readout is already 99.94 BLEU / 99.7% EM (dev 15.69, REC 14.64, PURE 26.32, gap +11.68); at step 2,828, 99.98 / 99.9% (dev 15.80, gap +11.49). The overshoot relative to the released evaluator’s step-1,820 readout of 78.8 is therefore not an artefact of training longer than the released run: under unclipped dynamics the training pool is essentially memorised before the released run’s best checkpoint.

**Multi-seed confirmation.** Across seeds 42–44 the unclipped protocol gives dev 15.74–16.32, REC 14.32–15.96, gaps +11.19/+11.31/+10.31, and readouts 99.98/95.25/99.98 BLEU (90.5–99.9% EM): the reproduction of both properties is stable across seeds, and every seed’s gap is within 1 BLEU point of the released +10.24.

**Interpretation and caveats.** The readout signature and the probe response are clipping-regime properties of the released recipe, constructible by disabling clipping (or, at intermediate strength, dialled to the released magnitude: clip 5.0 gives readout 82.69/68.4% vs. the released 78.8/70.7%, the closest single match of the ladder). The released operating point is bracketed but not matched in all coordinates—no single threshold places dev 13.38, readout 78.8, REC 12.78, and gap +10.24 simultaneously—so at least one further quantitative difference (CTC normalisation, stopping point, framework version, or data-order effects) persists; because the released config implies no clipping, the mismatch indicates the released run’s trajectory was slower than ours in dev-competence terms, not that it was clipped. The within-regime readout–gap association of the canonical panel ( $\rho=-0.56$ ; Sup. D) must not be extrapolated across regimes: across the ladder it is strictly positive (readout and gap rise together at every step from clip 1.0 to none).

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**Table S3:** Evidence-graded analysis registry. Rows are grouped by the role each analysis plays in the argument, not by when it was added.

| Evidence role                              | Analysis                                                      | Status | Multiplicity handling                                                                               |
|--------------------------------------------|---------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------|
| <i>Selected (descriptive)</i> <i>probe</i> | PURE-REC gap, original evaluator (post-selection)             | D      | descriptive MC estimate $p \leq 10^{-4}$ ; post-selection—see probe multiplicity table in main §4.1 |
| <i>Confirmatory</i>                        | 605-query donor-pool factorial (cells, effects, interaction)  | C      | Holm across 3 contrasts                                                                             |
|                                            | Template-holdout, BT-retrain, cross-fit                       | C      | paired bootstrap CIs                                                                                |
|                                            | Protocol checks (scorer, FPS, decoding sweep)                 | C/D    | exact-match / descriptive                                                                           |
|                                            | Frozen-protocol confirmation (seeds 1506/1607)                | C      | confirms family-side pattern                                                                        |
| <i>Triangulating</i>                       | 14 reconstruction gaps + prediction interval                  | C/D    | DiD CIs; PI descriptive                                                                             |
| (Finding 1: non-replication)               | Cross-metric gap table                                        | D      | direction/sign only                                                                                 |
|                                            | Conditioning controls (retrievers, gloss, random)             | D      | descriptive                                                                                         |
|                                            | Competence gate, trajectories, Pareto                         | D      | gate counts; 12-threshold grid                                                                      |
|                                            | Knowledge-distillation ladder + per- $\alpha$                 | D      | 15 with gap + dev                                                                                   |
| <i>Triangulating</i>                       | Reference replacement (single/dual); overlap strata           | D      | paired bootstrap of gap difference                                                                  |
| (Finding 2: reference sensitivity)         | Asymmetric floor-effect control                               | D      | BLEU-4/BLEU-1/chrF drops                                                                            |
|                                            | Evaluator $\times$ reference (7 $\times$ 3)                   | D      | descriptive decomposition                                                                           |
| <i>Triangulating</i>                       | Five diagnostic views (readout-with-generalization)           | D      | triangulation, not independent                                                                      |
| (Finding 3: readout signature)             | In-sample readout diagnosis (train/dev/test)                  | D      | no identity claim beyond train.pt                                                                   |
|                                            | Leakage sanity checks (permutation, equivariance)             | D      | exact-match / permutation                                                                           |
| <i>Robustness</i>                          | Donor-pool substitution; canonical B/C/D path                 | D      | query-index bootstrap CIs                                                                           |
|                                            | Alternative paths, Shapley, S4 system; balance/caliper/signer | D      | path-sensitivity display                                                                            |
|                                            | Rescue families (20); big-arch; long-schedule                 | D      | competence attempts                                                                                 |
|                                            | Released-checkpoint perturbation sweep                        | D      | lockstep collapse with competence                                                                   |
|                                            | IPW pool-origin robustness                                    | D      | balance robustness check                                                                            |
|                                            | Rank stability (9 systems $\times$ 7 evaluators)              | D      | $\tau_b/\rho$ /flip means                                                                           |
|                                            | Competence extrapolation (OLS/GP)                             | D      | non-IID; LOO CV (5 folds)                                                                           |
|                                            | Per-seed extension cells (8 seeds)                            | D      | descriptive; PI only                                                                                |
|                                            | Oracle/white-box transfer matrix                              | D      | permutation check                                                                                   |
|                                            | Template-density strata; donor-exclusion $\tau$ -sweeps       | D      | descriptive                                                                                         |
|                                            | CSL-Daily readout-ratio check                                 | D      | second-dataset boundary                                                                             |
|                                            | Transcript-slot diagnostic + extractor audit                  | D      | extractor P/R by system                                                                             |
| <i>Administrative</i>                      | Ethics/licensing declarations                                 | —      | updated each revision                                                                               |



**Table S4:** Canonical checkpoint registry. 76 training runs across 12 families (disk-scanned; auto-generated from `results/canonical_checkpoint_registry.json`). Of these 76 runs, 74 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry. The 74 decoded runs reduce to 73 unique weight binaries: the long-schedule seed-202 artifact is byte-identical to the validation-freq-misread seed-202 artifact (same wandb run, identical per-step losses, identical best-checkpoint SHA-256) and counts once. 69 non-degenerate decoded training runs enter the non-replication count (all 69 with strictly negative gaps; all share the clip=1.0 inference, resolved in Sup. U). Each family row’s columns sum to the Total row.

| Family                                 | Runs | Trained | Has gap<br>(beam-3) | Has dev | Enters                                                          |
|----------------------------------------|------|---------|---------------------|---------|-----------------------------------------------------------------|
| Faithful (train_faithful.py, 42–49)    | 8    | 8       | 8                   | 8       | RQ1 primary; non-reproduction (clip=1.0 panel; resolved Sup. U) |
| Reconstructions (primary)              | 6    | 6       | 6                   | 6       | multiseed, bootstrap, dose-resp.                                |
| Reconstructions (extension)            | 8    | 8       | 8                   | 8       | prediction interval, dose-resp.                                 |
| Validation-freq-misread                | 4    | 4       | 4                   | 4       | dose-resp.                                                      |
| Step-corrected (re-trained)            | 10   | 10      | 10                  | 10      | recipe-faithfulness; dose-resp.                                 |
| Joint-loss (greedy, 1801–1808)         | 8    | 8       | 8                   | 8       | protocol-robustness                                             |
| Joint-loss (beam-3, 1901–1908)         | 8    | 8       | 8                   | 8       | protocol-robustness                                             |
| Joint-loss supplementary (1809/1810)   | 1    | 1       | 0                   | 1       | exploratory; no canonical gap                                   |
| Train-pool ladder                      | 4    | 4       | 4                   | 4       | dose-resp., gap ranges                                          |
| Rescue (dropout/wd; only wd0 archived) | 1    | 1       | 0                   | 1       | gate                                                            |
| Confirmation                           | 2    | 2       | 2                   | 2       | dose-resp., family-side confirm.                                |
| Long-schedule                          | 1    | 1       | 1                   | 1       | gate; best seed dose-resp.                                      |
| Distillation students                  | 15   | 15      | 15                  | 15      | ladder, dose-resp.                                              |
| Total                                  | 76   | 76      | 74                  | 76      |                                                                 |
| Released original                      | —    | —       | 1                   | 1       | audit target (not trained)                                      |

Per-family columns sum to the Total row. Has-gap counts reflect the disk-scanned registry: 74 beam-3-decoded training runs; one validation-freq-misread seed-202 / long-schedule collision reduces unique binaries to 73. Five of the 74 decoded runs are degenerate (three  $\alpha=1.0$  distillation students with empty output and two smallest ladder fractions with BLEU $\approx 0$ ), leaving 69 non-degenerate runs that enter the headline non-replication count. The large-arch (4), rescue-lr (8), rescue-expanded (11 of 12), cross-fit (2), and BT-retrained-holdout (1) families listed in earlier registries were not recoverable from disk after the 2026-07-14 host compromise documented in our security incident report; their family-level summary statistics had been retained in the manual legacy registry but cannot be reproduced from current artifacts, and they are omitted from this disk-scanned table. The 11 released-weight fine-tune variants (Gaussian noise, fine-tuning) live under `checkpoints/finetune_released/` and are NOT counted as from-scratch training.

**Table S5:** Spearman correlations across the current uniform-readout set (38 constructible plus the released evaluator; 37 constructible have canonical gaps). “Non-degenerate” excludes the three  $\alpha=1.0$  distillation students that decode to empty output. Dev-based pairs use the checkpoints that also have uniform dev decodes. Supersedes the 31-point table of earlier rounds (`results/readout_gap_correlation.json`).

| Pair                                       | $\rho$ | $p$                  | $n$ |
|--------------------------------------------|--------|----------------------|-----|
| Readout vs. dev BLEU-4 (incl. released)    | 0.880  | $1.5 \times 10^{-6}$ | 18  |
| Readout vs. dev BLEU-4 (excl. released)    | 0.858  | $1.1 \times 10^{-5}$ | 17  |
| Readout vs. PURE-REC gap (incl. released)  | -0.444 | 0.0053               | 38  |
| Readout vs. PURE-REC gap (excl. released)  | -0.564 | 0.00028              | 37  |
| Readout vs. gap (non-deg., incl. released) | -0.319 | 0.062                | 35  |
| Readout vs. gap (non-deg., excl. released) | -0.438 | 0.0095               | 34  |
| Gap vs. dev BLEU-4 (incl. released)        | 0.502  | 0.040                | 17  |
| Readout vs. train NLL                      | -0.561 | 0.030                | 15  |
| Readout vs. train token accuracy           | 0.509  | 0.052                | 15  |

**Table S6:** Path sensitivity of the donor-pool substitution decomposition under the original evaluator (common 622-query support; 10,000-resample query-index bootstrap CIs). S1=SEEN-PURE (7,060-pool, max-Jaccard), S2=SEEN-RAND640 (640-pool, max), S3=SEEN-MATCHED (7,060-pool, Jaccard-matched), S4=SEEN-RAND640-MATCHED (640-pool, matched), U=UNSEEN-PURE (test pool). Every ordering telescopes to the same total ( $A$ ) = S1-U = +14.7.

| Ordering                 | Step           | Size contrast     | Selection contrast | Origin contrast   |
|--------------------------|----------------|-------------------|--------------------|-------------------|
| Canonical (prespecified) | S1→S2→S3→U     | +7.2 [+5.9, +8.4] | -0.8 [-1.8, +0.2]  | +8.4 [+7.3, +9.4] |
| Alternative 1            | S1→S2→S4→U     | +7.2 [+5.9, +8.4] | +1.8 [+1.1, +2.5]  | +5.8 [+4.7, +6.8] |
| Alternative 2            | S1→S3→S4→U     | +2.6 [+1.7, +3.5] | +6.4 [+5.1, +7.7]  | +5.8 [+4.7, +6.8] |
| Shapley average          | over orderings | +4.9 [+4.0, +5.8] | +4.1 [+3.3, +4.9]  | +5.8 [+4.7, +6.8] |

**Table S7:** Factorial design and balance (605 targets). SMD = standardized mean difference, seen vs unseen within stratum.

| Quantity                       | seen_high    | seen_low | unseen_high | unseen_low |
|--------------------------------|--------------|----------|-------------|------------|
| LaBSE similarity (mean)        | 0.813        | 0.654    | 0.776       | 0.622      |
| Duration (frames, mean)        | 100.2        | 99.5     | 99.2        | 98.8       |
| Word count (mean)              | 13.3         | 13.6     | 13.3        | 13.4       |
| Same-signer rate               | 100%         | 100%     | 100%        | 100%       |
| Similarity SMD (seen - unseen) | +0.41 (high) |          | +0.38 (low) |            |

**Table S8:** Full  $2 \times 2$  exposure factorial (605 targets; corpus sacreBLEU in BLEU points; NLL per token). CIs: 10,000-resample bootstrap (original) and seed-level  $t$  across six reconstructions; Holm correction across three contrasts.

| Metric | Evaluator    | seen_hi | seen_lo | unseen_hi | unseen_lo | seen main                    | sim. main             | inter.                       |
|--------|--------------|---------|---------|-----------|-----------|------------------------------|-----------------------|------------------------------|
| BLEU   | Original     | 11.12   | 2.31    | 7.19      | 2.08      | <b>+2.08</b><br>[+1.6, +2.6] | +6.96<br>[+5.9, +8.1] | <b>+3.70</b><br>[+2.6, +4.8] |
| BLEU   | Seeds (mean) |         |         |           |           | +0.51                        | +4.06                 | -0.39                        |
| NLL    | Original     | 3.74    | 4.30    | 3.93      | 4.37      | -0.128                       | -0.498                | -0.124                       |
| NLL    | Seeds (mean) |         |         |           |           | -0.064                       | -0.179                | -0.004                       |

**Table S9:** Multivariate donor balance (622-query common support). All  $|SMD| < 0.12$ .

| Covariate            | SEEN-MATCHED mean | UNSEEN mean | SMD    | Per-query overlap            |
|----------------------|-------------------|-------------|--------|------------------------------|
| Source-text Jaccard  | 0.411             | 0.415       | -0.021 | $ \Delta J  \leq 0.05$ : 96% |
| LaBSE cosine         | 0.782             | 0.774       | +0.068 | —                            |
| Duration (s)         | 3.76              | 3.74        | +0.014 | $ \Delta  \leq 1$ s: 57%     |
| Word count           | 12.29             | 12.31       | -0.003 | $ \Delta  \leq 2$ : 69%      |
| Template density     | 0.310             | 0.326       | -0.075 | —                            |
| Same signer as query | 0.183             | 0.228       | -0.111 | both same-signer: 4.7%       |

**Table S10:** Released `config.yaml` load-bearing field-by-field mapping: original-framework semantics (empirically pinned by the released `validations.txt` where noted) vs. our v1–v3 reconstruction semantics (with defects) vs. our faithful v4 semantics. Unlisted fields (model architecture, dropout, initialisers, max lengths) are applied verbatim and identical across v1–v4.

| Field (value)                                                                                    | (released) | Original semantics                                                                                                                                                                                                                                                                          | framework                         | v1–v3 (defect)                                                                                                                                                | semantics                                                                                                                                                                                                                         | v4 faithful semantics |
|--------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>translation_normalization</code> (batch)                                                   |            | batch-sum CE ÷ sentences; logged dev Translation Loss is the raw token sum (43,440.8 = 7,813 × ln 259.84)                                                                                                                                                                                   |                                   | CE reduction="mean" (per-token): ~15–30× smaller gradient scale; with Adam weight-decay coupling this changes effective regularization                        | CE reduction="sum" ÷ batch sentences; smoke-run step-14 dev token-sum 42,981.5 vs. released 43,440.8 (1.1% off)                                                                                                                   |                       |
| dev NLL token count                                                                              |            | counts (mask valid)                                                                                                                                                                                                                                                                         | <i>valid</i> (mask True = target) | ( <code>txt_mask.sum()</code> counts <i>padding</i> positions (all of v1–v3); also truncates v2 greedy-selection hypotheses/references by padding count)      | <code>txt_mask.sum()</code> ; and per-token NLL computed on valid tokens only                                                                                                                                                     | PPL and NLL           |
| <code>recognition_loss_weight</code> (1.0)                                                       |            | CTC+translation; logged Recognition Loss column normalisation not exactly identifiable (17.28 at step 14 matches neither per-sentence 39.9 nor per-gloss-token 12.3 in our stack)                                                                                                           |                                   | v1: CTC omitted entirely; step-corrected family: <code>--rec-weight 0</code> (CTC off); v2/v3: joint with per-token-mean CTC                                  | joint CTC+translation, CTC token-sum ÷ sentences (batch-consistent); recognition-column normalisation documented as not identifiable                                                                                              |                       |
| <code>validation_freq</code> (14)                                                                |            | every 14 <i>optimiser steps</i> (logged Steps: 14, 28, ...)                                                                                                                                                                                                                                 |                                   | v1: misread as 14 <i>epochs</i> ; v1.5+/v2/v3: 14 steps (corrected)                                                                                           |                                                                                                                                                                                                                                   | 14 optimiser steps    |
| <code>eval_translation_beam_size</code> (3) for selection                                        |            | beam-size; decode for validation BLEU (WER/BLEU columns of released log)                                                                                                                                                                                                                    | beam-3                            | v1: selection branch never saved (hard-coded <code>improved=False</code> ); v1.5/v2: teacher-forced greedy argmax, not autoregressive; v3: beam-3 (corrected) | autoregressive beam-3 (via <code>back_translate</code> ), identical to the evaluation decoder                                                                                                                                     |                       |
| <code>epochs</code> (3000) / stopping                                                            |            | released run ended at step 2828 (101 epochs) after 72 consecutive non-improving validations—patience-5 early stop is <i>provably not</i> the released rule; LR-floor is the only config-reproducible stop                                                                                   |                                   | 300-epoch cap with early-stop patience 15                                                                                                                     | 3000-epoch cap; default stop rule = LR floor (1e-8); patience rule available but off and documented as not released-faithful                                                                                                      |                       |
| <code>scheduling</code> (plateau), <code>decrease_factor</code> (0.8), <code>patience</code> (5) |            | ReduceLROnPlateau(mod <i>present</i> ) since v1, but on dev BLEU; every logged decay fires on the 6th consecutive non-improving validation                                                                                                                                                  |                                   | reached only ~60–90 validations before the patience-15 stop                                                                                                   | identical scheduler wiring; LR logged after <code>scheduler.step</code> as in the released log                                                                                                                                    |                       |
| <code>batch_size</code> (256)                                                                    |            | 256 sentences per optimiser step                                                                                                                                                                                                                                                            |                                   | v1: physical 64 × grad-accum 4 (inferred); v1.5+/v2/v3: 256                                                                                                   | 256                                                                                                                                                                                                                               |                       |
| <code>random_seed</code> (42)                                                                    |            | 42 (released run)                                                                                                                                                                                                                                                                           |                                   | never used as the primary seed (101–1405, 1701–1908 sweeps)                                                                                                   | 42 is the primary run; 43–49 as within-implementation dispersion                                                                                                                                                                  |                       |
| gradient clipping                                                                                |            | <b>not specified in config</b> ⇒ signjoey framework default <b>no clipping</b> (build <code>gradient_clipper</code> returns <code>None</code> when <code>clip_grad_val/clip_grad_norm</code> absent; framework clips by value, not norm)—resolved in round 33 as the eighth defect (Sup. U) |                                   | hard-coded <code>max_norm=1.0</code> (with token-mean loss it almost never binds)                                                                             | <code>--clip 1.0</code> default, disclosed as author inference; with batch-normalised loss the norm genuinely binds (first steps 377/568/145)— <i>overturned</i> : unclipped reproduces the signature and probe response (Sup. U) |                       |

Fields applied verbatim in all our versions (not shown): `learning_rate` 0.001, `betas` [0.9, 0.998], `weight_decay` 0.001, `label_smoothing` 0.0, `learning_rate_min` 1e-8, `keep_last_ckpts` 1, architecture block (transformer, 256/512/8/3, dropout 0.1, softsign, xavier), `eval_translation_beam_alpha` -1, `translation_max_output_length` 30, `skeleton_subsample` 2, `txt_lowercase` true, `max_sent_length` 400. Remaining author inferences: exact CTC reduction of the original stack (signjoey applies the raw sum; not identifiable from the released log); embeddings `norm_type`

**Table S11:** Epoch-validation approximation (validate every 14 epochs, not steps): 300 epochs, dev-NLL selection. All four seeds produce negative PURE-REC gaps under the canonical §3.2 donor registry; none shows the reversal or readout signature.

| Seed               | ep.   | dev NLL | REC   | PURE gap |
|--------------------|-------|---------|-------|----------|
| 101                | 27    | 2.11    | 10.12 | -0.60    |
| 202                | —     | —       | 10.14 | -1.70    |
| 303                | —     | —       | 8.74  | -1.51    |
| 404                | —     | —       | 8.43  | -0.52    |
| Released evaluator | 1,820 | —       | 12.78 | +10.24   |

**Table S12:** Pose-perturbation battery (corpus sacreBLEU, BLEU points) under the released evaluator and the best rescue checkpoint (wd0, seed 202). The TN-PURE-v1 identity column here (23.79) reflects the pre-exclusion materialization used at the time this battery was decoded; the canonical materialization gives 23.02 (gap +10.24). Because the battery varies the pose and re-decodes, not the donor choice, the perturbation differentials are unaffected by the materialization difference (Sup. N: mean  $|\Delta\text{gap}| = 0.65$ , max 1.45).

| Evaluator      | System     | identity | reversal | window shuffle | jitter 5% | hand mask | face mask |
|----------------|------------|----------|----------|----------------|-----------|-----------|-----------|
| Original       | REC-v1     | 12.78    | 7.25     | 11.97          | 12.51     | 1.35      | 3.15      |
| Original       | TN-PURE-v1 | 23.79    | 7.72     | 18.35          | 23.91     | 1.30      | 4.06      |
| wd0 (seed 202) | REC-v1     | 10.73    | 6.26     | 10.54          | 10.60     | 0.63      | 1.58      |
| wd0 (seed 202) | TN-PURE-v1 | 10.98    | 6.66     | 9.65           | 10.92     | 0.58      | 1.43      |

**Table S13:** Cluster-aware bootstrap for the headline PURE-REC gap (canonical §3.2 registry, original refs, released-evaluator hypotheses from `results/gap_43_canonical_beam3.items/released_{gt,pure}.json`). All CIs are 10,000-resample sacreBLEU `tok:13a` exp-smooth over precomputed per-item n-gram sufficient statistics. With so few clusters (especially show with only 2), these are *exploratory sensitivity checks*, not confirmatory tests. Donor reuse: 641 queries select 565 unique donors (61 donors serve more than one query; maximum reuse 6; Gini 0.11).

| Analysis                    | Design   | Clusters | Gap    | 95% CI          |
|-----------------------------|----------|----------|--------|-----------------|
| Query-level (baseline)      | item     | 641      | +10.24 | [+8.88, +11.62] |
| Donor cluster               | donor    | 565      | +10.25 | [+8.87, +11.64] |
| Donor two-way (query×donor) | nested   | 565      | +10.25 | [+8.79, +11.74] |
| Signer                      | signer   | 9        | +10.30 | [+9.29, +11.39] |
| Broadcast show              | show     | 2        | +9.27  | [+5.16, +11.44] |
| Broadcast date              | date     | 297      | +10.25 | [+8.83, +11.72] |
| Signer×Show                 | combined | 17       | +10.14 | [+7.89, +11.91] |

**Table S14:** PHX-public pure-donor source-text donor-exclusion  $\tau$ -sweep under the original SLRTP2025 BT (ical tie-breaking, 12.5 fps; sacreBLEU on the 0–100 scale). The retrieval-exceeds-REC reversal survives exact exclusion and  $\tau=0.5$  exclusion.

| Condition                                                                          | sacreBLEU | $\Delta$ vs REC | Rev |
|------------------------------------------------------------------------------------|-----------|-----------------|-----|
| REC baseline                                                                       | 12.8      | —               |     |
| Text-nearest pure (no exclusion; pre-exclusion historical materialization, Sup. N) | 23.8      | +11.0           |     |
| Text-nearest pure (canonical, w/ exclusion)                                        | 23.02     | +10.24          |     |
| Exact-text excluded                                                                | 23.0      | +10.2           |     |
| $\tau=0.7$ (Jaccard > 0.7 excluded)                                                | 21.2      | +8.4            |     |
| $\tau=0.5$ (Jaccard > 0.5 excluded)                                                | 17.8      | +5.0            |     |
| $\tau=0.3$ (Jaccard > 0.3 excluded)                                                | 8.0       | −4.8            |     |
| $\tau=0.1$ (Jaccard > 0.1 excluded)                                                | 0.2       | −12.5           |     |
| Random donor (mean of 20 seeds)                                                    | 0.9       | −11.6           |     |

**Table S15:** Template-density stratification (sacreBLEU 0–100). Retrieval exceeds REC at all density levels.

| Density stratum | Mean density | REC  | Retrieval | Gap   |
|-----------------|--------------|------|-----------|-------|
| Low (n=249)     | 0.152        | 5.9  | 16.6      | +10.7 |
| Mid (n=179)     | 0.254        | 9.5  | 19.7      | +10.1 |
| High (n=213)    | 0.510        | 26.1 | 38.6      | +12.4 |

**Table S16:** Holdout boundary controls (sacreBLEU, 0–100; historical—per-item artifacts retained, training runs unrecoverable after the 2026-07-14 host incident). None reproduces the canonical +10.24 reversal.

| Control                       | Setup                               | REC  | Retrieval | Gap    |
|-------------------------------|-------------------------------------|------|-----------|--------|
| Same evaluator                | test split (template-overlap)       | 12.8 | 23.02     | +10.24 |
| Same evaluator                | template-holdout (seen in training) | 77.0 | 20.1      | −56.9  |
| BT-retrained (unseen holdout) | train-core 6,535                    | 6.2  | 5.4       | −0.9   |
| Retrained BT on PHX-public    | unseen to both                      | 8.7  | 8.3       | −0.4   |
| Cross-fit A/B (donor-unseen)  | pooled, $n=659$                     | 4.0  | 3.9       | −0.1   |

**Table S17:** Frozen template-holdout controls (1,538 windows,  $\alpha=0.5$ ; sacreBLEU 0–100).

| System                                 | sacreBLEU                 | Empty %    | Mean Jac.           |
|----------------------------------------|---------------------------|------------|---------------------|
| REC baseline                           | 13.10                     | 9.95       | —                   |
| Random train donor                     | 0.04                      | 11.83      | —                   |
| Text-nearest train donor               | 2.30                      | 10.66      | —                   |
| LaBSE / multi-SBERT (donor copy)       | 1.73 / 1.51               | 9.2 / 11.3 | —                   |
| Noisy-gloss retrieval (strong NMT)     | 0.40                      | 10.90      | —                   |
| Retrieval composition ( $\alpha=0.5$ ) | 2.20                      | 10.79      | —                   |
| $\tau=0.1$ / 0.3 / 0.5 / 0.7 exclusion | 0.04 / 1.05 / 1.93 / 2.01 | 10.0–10.5  | 0.10/0.29/0.40/0.42 |
| BM25 / TF-IDF nearest                  | 1.75 / 2.36               | 10.8 / 8.8 | —                   |

**Table S18:** Competence extrapolation: gap predictions at dev BLEU-4 = 13.38 from three fits over 65 unique non-released binaries (dev BLEU-4 0–10.5; 61 non-degenerate). All predictions are far below the observed canonical +10.24. PIs are *descriptive* (non-IID checkpoints); they should not be read as calibrated uncertainty intervals.

| Fit                                      | R <sup>2</sup> / LML | Prediction at 13.38 | 95% PI (descriptive) |
|------------------------------------------|----------------------|---------------------|----------------------|
| OLS linear                               | 0.48                 | −1.84               | [−2.67, −1.01]       |
| OLS quadratic                            | 0.50                 | −1.09               | [−1.83, −0.35]       |
| Gaussian Process (RBF + white)           | −48.1                | −1.00               | [−2.16, +0.17]       |
| Released evaluator (observed, canonical) | —                    | +10.24              | [+8.88, +11.62]      |

**Table S19:** Leave-one-family-out cross-validation of the competence extrapolation. Each row refits all three models after removing one checkpoint family (ladder fractions are grouped with the distillation family, matching the §4.2 figure family mapping). The GP upper bound stays below +0.3 across all folds; OLS quadratic is the most sensitive (shifts sign when distillation is removed). All intervals descriptive.

| Family removed              | $n$ remaining | OLS-lin pred | OLS-quad pred | GP pred | GP upper | OLS-lin R <sup>2</sup> |
|-----------------------------|---------------|--------------|---------------|---------|----------|------------------------|
| None (full)                 | 42            | −1.84        | −1.09         | −1.00   | +0.17    | 0.48                   |
| Reconstructions (14)        | 28            | −1.84        | −2.07         | −0.89   | +0.25    | 0.57                   |
| Distillation (19)           | 23            | −0.78        | +0.23         | −1.18   | −0.41    | 0.03                   |
| Validation-freq-misread (8) | 34            | −1.89        | −1.32         | −0.97   | +0.22    | 0.56                   |
| Rescue (1)                  | 41            | −1.82        | −0.97         | −0.98   | +0.18    | 0.47                   |

**Table S20:** CSL-Daily Mode B donor-exclusion (1,176 sequences; 0–100 scale).

| Condition            | sacreBLEU | BLEU-1 |
|----------------------|-----------|--------|
| REC baseline         | 26.7      | 79.1   |
| Text-nearest donor   | 1.6       | 43.5   |
| Random donor         | 0.0       | 3.9    |
| $\tau=0.5$ exclusion | 1.0       | 42.6   |
| BM25 retrieval       | 1.5       | 39.2   |
| TF-IDF retrieval     | 1.2       | 38.5   |

**Table S21:** Label permutation test (1,000 items, reference labels shuffled *after* decoding). BLEU collapses from 78.6 to  $\sim 1.0$ , confirming output–reference alignment. This test alone cannot distinguish pose-conditioned output from ID/cache-based retrieval; the input-side pose permutation below addresses that.

| Condition                  | BLEU  | EM    |
|----------------------------|-------|-------|
| Original (correct pairing) | 78.61 | 0.701 |
| Permutation 0              | 1.06  | 0.000 |
| Permutation 1              | 1.30  | 0.000 |
| Permutation 2              | 0.90  | 0.002 |

**Table S22:** Input-side pose permutation with output-equivariance (100 items). Pose tensors are shuffled *before* decoding while IDs stay fixed. The equivariance column verifies  $E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$ .

| Condition                         | BLEU vs slot ref | Detail                |
|-----------------------------------|------------------|-----------------------|
| Original decode vs own reference  | 75.62            | baseline              |
| Shuffled poses vs slot reference  | 0.95             | BLEU collapses        |
| Shuffled poses vs donor reference | 75.62            | identical to original |
| Output-equivariance match rate    | —                | 100/100 (100%)        |
| Donor-length match rate           | —                | 100/100 (100%)        |
| Slot-length match rate            | —                | 5/100 (5%)            |



**Table S23:** Stratified exact-match rate on the full 7,060-item training pool under beam=3 free-decode.

| Stratum                            | $n$   | EM    |
|------------------------------------|-------|-------|
| <b>By signer</b>                   |       |       |
| Signer01                           | 1,852 | 0.683 |
| Signer02                           | 83    | 0.590 |
| Signer03                           | 579   | 0.736 |
| Signer04                           | 1,049 | 0.748 |
| Signer05                           | 1,623 | 0.701 |
| Signer06                           | 35    | 0.714 |
| Signer07                           | 762   | 0.696 |
| Signer08                           | 844   | 0.715 |
| Signer09                           | 233   | 0.734 |
| <b>By reference length</b>         |       |       |
| Short ( $\leq 5$ words)            | 203   | 0.837 |
| Medium (6–10)                      | 1,319 | 0.685 |
| Long (11–20)                       | 4,335 | 0.714 |
| Very long ( $>20$ )                | 1,203 | 0.686 |
| <b>By text repetition in train</b> |       |       |
| Unique (1 occurrence)              | 6,777 | 0.702 |
| Low (2–3)                          | 57    | 0.737 |
| Mid (4–10)                         | 76    | 0.684 |
| High ( $>10$ )                     | 150   | 0.940 |

**Table S24:** Same-signer held-out control: train (seen) vs. test (unseen) free-decode. The 67-point BLEU gap is consistent with training-pool exposure contributing to the readout, though the lack of full covariate matching means we cannot quantify how much of the gap is exposure vs. distributional shift.

| Split         | $n$   | BLEU  | EM    | Shared signers |
|---------------|-------|-------|-------|----------------|
| Train (seen)  | 7,060 | 78.82 | 0.707 | 9              |
| Test (unseen) | 500   | 12.19 | 0.032 | 9 (same)       |

**Table S25:** Three-way data provenance for PHX-public. The released evaluator’s training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs carry the missingness label `absent_from_slrtp2025_released_pose_records`—a status flag rather than an explained mechanism (SI Sup. A). SHA-256 hashes verify file integrity.

| Split source                                         | Train | Dev | Test |
|------------------------------------------------------|-------|-----|------|
| Official PHOENIX-2014T                               | 7,096 | 519 | 642  |
| Released SLRTP2025 pose (.pt)                        | 7,060 | 515 | 641  |
| Released evaluator <code>config.yaml</code> manifest | 7,060 | 515 | 641  |
| Missing from official                                | 36    | 4   | 1    |

**Table S26:** Checkpoint accounting, auto-generated from `results/accounting-table.json` (single source of truth, disk-scanned). “Runs” = from-scratch training executions; “Decoded” = beam-3 decoded on the canonical PHX-public gap panel; “Unique” = distinct SHA-256 weight binaries; “Non-deg.” = non-degenerate (BLEU > 0, excluding 3  $\alpha=1.0$  distillation students with empty output and 2 smallest ladder fractions). The partially-corrected tier (14 corrected-recipe runs) provides recipe-faithfulness evidence; secondary (14 legacy, of which 6 prospectively specified) provides the prospectively specified non-reproduction direction plus stability; diagnostic is mechanism exploration.

| Tier / Family                                   | Runs      | Decoded   | Unique    | Non-deg.  |
|-------------------------------------------------|-----------|-----------|-----------|-----------|
| <b>Faithful</b>                                 | 8         | 8         | 8         | 8         |
| Faithful<br>(train_faithful.py,<br>seeds 42–49) | 8         | 8         | 8         | 8         |
| <b>Partially-corrected</b>                      | 14        | 14        | 14        | 14        |
| Validation-freq-misread                         | 4         | 4         | 4         | 4         |
| Step-corrected<br>(re-trained)                  | 10        | 10        | 10        | 10        |
| <b>Secondary</b>                                | 14        | 14        | 14        | 14        |
| Paper-derived<br>(prospectively specified)      | 6         | 6         | 6         | 6         |
| Paper-derived<br>(extension seeds)              | 8         | 8         | 8         | 8         |
| <b>Diagnostic</b>                               | 40        | 38        | 37*       | 33        |
| Joint-loss (greedy selection)                   | 8         | 8         | 8         | 8         |
| Joint-loss (beam-3 selection)                   | 8         | 8         | 8         | 8         |
| Joint-loss<br>(supplementary, 1809/1810)        | 1         | 0         | 0         | 0         |
| Confirmation                                    | 2         | 2         | 2         | 2         |
| Rescue                                          | 1         | 0         | 0         | 0         |
| (dropout/wd variants)                           |           |           |           |           |
| Distillation                                    | 15        | 15        | 15        | 12        |
| Ladder (data fractions)                         | 4         | 4         | 4         | 2         |
| Long-schedule                                   | 1         | 1         | 0*        | 1         |
| <b>Total trained</b>                            | <b>76</b> | <b>74</b> | <b>73</b> | <b>69</b> |

\*One validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run); it counts once in the unique-binary column (total unique = 74 decoded – 1 collision = 73). The released evaluator (1 checkpoint) is not a training run and is not included above. Eleven released-weight fine-tune variants (Gaussian noise, local perturbation) are not from-scratch training and are not counted here. The previously-reported large-arch, rescue-lr, cross-fit, and B35-retrained holdout families were not recoverable from disk after a host compromise (SI Sup. D) and are omitted; their family-level summary statistics had been retained in the legacy manual registry but cannot be reproduced from artifacts.

**Table S27:** Novelty relative to closely related PHOENIX-2014T and learned-metric reproducibility studies.

| Study                  | Primary finding                                                                                                                    | Method                                                                                                      | Distinction from present work                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Czehmann et al. (2026) | Reference replace-ment changes BLEU scores and rankings on PHOENIX-2014T                                                           | Human back-translations of test set                                                                         | We apply the matched-subset protocol to the released evaluator with a frozen replay probe and quantify the paired reference-frame contrast |
| Alkain et al. (2026)   | Highest train-test text overlap among tested SLT corpora                                                                           | N-gram overlap quantification                                                                               | We use their finding to contextualize why PHX-public is particularly susceptible to reference coupling                                     |
| Artiaga et al. (2025)  | Signer-dependent evaluation overestimates SLT capability                                                                           | Signer-disjoint splits                                                                                      | We include signer as a regression covariate (not signer-disjoint evaluation; $\Delta R^2$ from removing the signer block is $\leq 1\%$ )   |
| Jiang et al. (2025)    | BT likelihood is a more stable correlate of human judgment than decoded BLEU                                                       | Meta-evaluation of pose/embedding/BT metrics                                                                | We report that teacher-forced likelihood does not reproduce the reversal (REC NLL 3.060 vs PURE 3.479)                                     |
| Chen et al. (2022)     | BERT-based metrics fail to reproduce due to missing preprocessing/code                                                             | Multi-implementation replication                                                                            | We extend the learned-metric reproducibility question to a full SLP evaluator pipeline with 68 training runs across 11 families            |
| This work              | Our disclosed reconstruction from the public artifacts did not attain the released evaluator’s competence or replay-probe behavior | 68 training runs across 11 families, fixed-output cross-evaluator re-decoding, reference/donor substitution | —                                                                                                                                          |

**Table S28:** Public-artifact provenance of each reconstruction family. “Dev BLEU-4” is the uniform full-pool beam-3 value (§3.4); the released evaluator scores 13.38. “Selection” is the checkpoint-selection rule. The faithful runs implement the released config exactly; the paper-derived runs use NLL selection on a legacy script; the step-corrected runs match the released `validation_freq=14` JoeyNMT semantics with decoded-BLEU selection; the joint-loss variants additionally correct the recognition-loss omission.

| Family                              | Public source used                                     | Known deviation                                             | Runs                            | Selection    | Dev BLEU-4   |
|-------------------------------------|--------------------------------------------------------|-------------------------------------------------------------|---------------------------------|--------------|--------------|
| <b>Faithful</b>                     | released config.yaml exactly                           | none (seven defects corrected; seed 42 first)               | 8                               | beam-3-BLEU  | 11.59–12.62  |
| Paper-derived (prosp.-spec. + ext.) | config.yaml + slt architecture                         | legacy script; epochs; NLL selection                        | 300 14                          | NLL on dev   | 6.5–9.4      |
| Validation-freq-misread             | config.yaml (decoded-BLEU)                             | <code>validation_freq=14</code> misread as epochs initially | 4                               | decoded BLEU | 8.3–9.8      |
| Step-corrected (re-trained)         | config.yaml JoeyNMT semantics                          | + fixed step interpretation; translation-only loss          | <code>validation_freq</code> 10 | greedy-BLEU  | 7.6–10.5     |
| Joint-loss (greedy)                 | config.yaml ( <code>recognition_loss_weight=0</code> ) | adds CTC; token-mean selection                              | 8                               | greedy-BLEU  | 6.3–10.2     |
| Joint-loss (beam-3)                 | config.yaml ( <code>eval_beam=3</code> )               | adds CTC; token-mean loss; beam-3 selection                 | 8                               | beam-3-BLEU  | 8.7–10.5     |
| Confirmation                        | recipe-faithful protocol                               | independent re-run for reproducibility                      | 2                               | decoded BLEU | 9.0–9.5      |
| Long-schedule                       | config.yaml (epochs=3000)                              | extended training budget                                    | 1                               | decoded BLEU | $\leq 10.02$ |
| Rescue (dropout/wd)                 | hyperparameter grid                                    | dropout/wd variations; none passed gate                     | 1                               | decoded BLEU | $\leq 10.02$ |
| Distillation                        | released checkpoint as teacher                         | student-teacher KL; students trained from scratch           | KL; 15                          | decoded BLEU | 7.0–9.0      |
| Ladder (data fractions)             | config.yaml                                            | data-subset fractions (0.125–0.75 $\times$ )                | 4                               | decoded BLEU | $\leq 9.8$   |
| Joint-loss (supplementary)          | exploratory v2 seeds (1809/1810)                       | out-of-protocol v2 explorations                             | 1                               | —            | not decoded  |
| <b>Total</b>                        |                                                        |                                                             | <b>76</b>                       |              |              |

**Table S29:** Teacher-forced and free-decode diagnostic statistics for the released evaluator and fourteen reconstruction seeds (full table in `results/checkpoint_stats.json`). Teacher-forced values on 500 randomly sampled items (seed 42, sampled without replacement; the sampling SD across 5 re-samples is  $\leq 0.3$  pp for token accuracy and  $\leq 0.01$  for NLL/tok); free-decode values are from the training pool (7,060 train / 515 dev) under the uniform beam-3 protocol (§3.4). sacreBLEU signature: `sacreBLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1` (max n-gram order 4, the default).

| Checkpoint              | Train tok. acc. | Train NLL/tok | Train free-decode BLEU | Dev NLL/tok | Dev free-decode BLEU |
|-------------------------|-----------------|---------------|------------------------|-------------|----------------------|
| Released                | 0.965           | 0.158         | 78.82                  | 3.236       | 13.8                 |
| Reconstructions (range) | 0.550–0.596     | 1.72–1.92     | 7.20–9.25              | 2.63–2.68   | 6.49–9.25            |
| Reconstructions (mean)  | 0.572           | 1.82          | 8.35                   | 2.66        | 8.35                 |

**Table S30:** Custom-terminology registry. Each term maps to its estimand, the dataset on which it is computed, and its evidence grade (all descriptive).

| Term                       | Estimand                                                    | Dataset                      | Grade              |
|----------------------------|-------------------------------------------------------------|------------------------------|--------------------|
| Training-pool read-out     | In-sample autoregressive free-decode BLEU/EM                | Full 7,060-item training set | D                  |
| PURE-REC gap               | Corpus sacreBLEU-4 difference (PURE – REC)                  | 641-item PHX-public test     | D (post-selection) |
| Operational tolerance gate | One-sided: dev BLEU-4 $\geq 12.38$ and dev WER $\leq 86.37$ | 515-item dev set             | D                  |

**Table S31:** Claim hierarchy. Each key claim mapped to its evidence tier and supporting evidence.

| Claim                                                                                   | Tier | Evidence                                                                                            |
|-----------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------|
| Our disclosed reconstruction did not attain released evaluator competence               | D    | 28 recipe-conditioned runs, all dev BLEU < 10.5; not a public-artifact-insufficiency claim          |
| Probe response does not reproduce across canonical (clip=1.0) constructible checkpoints | D    | 69/69 non-degenerate gaps negative; resolved by the clipping sensitivity (Sup. U: unclipped +11.19) |
| Probe response reflects a causal checkpoint-identity effect                             | NI   | Resolved as a clipping-regime property; no identity effect remains to attribute                     |
| Headline gap +10.24 is post-selection                                                   | S    | Max-gap probe among 8 explored; CI descriptive                                                      |
| Reference-frame contrast: gap falls 90.5% under human back-translations                 | S    | Single-annotator, 461-item, 4 confounded changes                                                    |
| Origin effect persists under matched pools (+9.56)                                      | S    | 148 items, SMD<0.1; association not causal                                                          |
| Probe is adversarial lexical-coupling stress test                                       | D    | Source = reference in PHX-public; retriever uses source Jaccard                                     |

**Table S32:** Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL does not reproduce it (SI Sup. O). Item-level pose-DTWp is not reported (see main text).

| System                       | BLEU-4       | BLEU-1       | chrF         | ROUGE-L      | WER          | BERTSc.      |
|------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| REC-v1 (recorded test poses) | 12.78        | 34.40        | 34.59        | 35.20        | 85.77        | 0.757        |
| PT-v1                        | 1.59         | 16.91        | 19.03        | 17.26        | 101.68       | 0.669        |
| TN-PTCOMP                    | 18.86        | 46.35        | 41.88        | 45.17        | 77.36        | 0.793        |
| TN-PURE                      | <b>23.02</b> | <b>55.43</b> | <b>47.82</b> | <b>53.07</b> | <b>71.87</b> | <b>0.822</b> |
| Gap PURE-REC                 | +10.24       | +20.94       | +12.93       | +17.87       | -13.90       | +0.066       |

**Table S33:** Asymmetric floor-effect control (full-641 sensitivity analysis). PURE loses a larger fraction of its REC across all metrics when references are replaced, so simple symmetric compression is insufficient; the does not discriminate between reference-text defects and training-distribution alignment.

| Metric                    | REC (orig)                     | REC (human) | REC drop     | PURE (orig) | PURE (human) | P    |
|---------------------------|--------------------------------|-------------|--------------|-------------|--------------|------|
| BLEU-4                    | 12.78                          | 5.66        | 7.11 (55.7%) | 23.02       | 6.52         | 16.5 |
| BLEU-1                    | 34.40                          | 29.28       | 5.12 (14.9%) | 55.43       | 32.35        | 23.0 |
| chrF                      | 34.59                          | 32.75       | 1.84 (5.3%)  | 47.82       | 35.02        | 12.8 |
| Dual-ref gap (orig+human) | BLEU-4: +11.50; BLEU-1: +20.03 |             |              |             |              |      |

**Table S34:** PURE-REC gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in the main-text multiseed factorial table). chrF, WER and ROUGE-L use the official SLRTP2025 implementations. WER and NLL are sign-flipped so positive always means PURE is better than REC. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.13 package, whose module `__version__` string reads 0.3.12; `lang=de`, `idf=False`, no baseline rescaling; model revision logged in the artifact); per-checkpoint BERTScore cells are in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

| Evaluator | $\Delta$ BLEU-4 | $\Delta$ BLEU-1 | $\Delta$ chrF | $\Delta$ WER <sup>†</sup> | $\Delta$ NLL <sup>†</sup> | $\Delta$ BERTSc. |
|-----------|-----------------|-----------------|---------------|---------------------------|---------------------------|------------------|
| Original  | +10.24          | +20.94          | +12.93        | +13.90                    | -0.419                    | +0.066           |
| Seed 101  | -0.80           | -1.23           | -1.44         | -2.95                     | -0.219                    | -0.005           |
| Seed 202  | -1.30           | -1.33           | -1.57         | -3.52                     | -0.233                    | -0.006           |
| Seed 303  | -1.85           | -1.99           | -2.26         | -1.53                     | -0.240                    | +0.001           |
| Seed 404  | -1.81           | -1.60           | -1.90         | -1.87                     | -0.194                    | -0.001           |
| Seed 505  | -1.28           | -1.53           | -1.59         | -3.41                     | -0.268                    | -0.004           |
| Seed 606  | -1.24           | -1.26           | -1.55         | -3.46                     | -0.169                    | -0.001           |

<sup>†</sup>WER and NLL are sign-flipped: positive means PURE is better than REC (lower WER, lower NLL).



**Table S35:** Public SLRTP2025 challenge leaderboard on ph14t-test (Walsh et al. Walsh et al. (2025), Table 1), computed under the same released BT evaluator the main text audits; the “Ground Truth” row matches the canonical REC exactly. USTC-MoE is roughly even with REC on lexical BT metrics (BLEU-4  $-0.72$ , BLEU-1  $+0.45$ , chrF  $+2.24$ , WER  $+7.72$ ) despite a much larger DTW-MJE (0.0448) and duration ratio (1.438). This single-evaluator co-occurrence is descriptive and does not establish evaluator retrieval-favoring bias.

| System                        | BLEU-1       | BLEU-4       | chrF         | ROUGE        | WER↓   | DTW-MJE↓ |
|-------------------------------|--------------|--------------|--------------|--------------|--------|----------|
| Recorded test poses (REC)     | 34.40        | 12.78        | 34.59        | 35.20        | 85.77  | 0.0000   |
| Team 1 – USTC-MoE (retrieval) | <b>34.85</b> | <b>12.06</b> | <b>36.83</b> | <b>36.59</b> | 93.49  | 0.0448   |
| Team 3 – Hacettepe            | 30.44        | 9.59         | 29.70        | 30.64        | 88.88  | 0.0423   |
| Progressive Transformer       | 22.17        | 5.43         | 24.13        | 21.98        | 101.45 | 0.0418   |
| Team 2 – hfut-lmc             | 16.96        | 2.05         | 25.88        | 19.77        | 147.85 | 0.0403   |
| TN-PURE-v1 (replay probe)     | 55.43        | 23.02        | 47.82        | 53.07        | 71.87  | 0.00678  |

**Table S36:** Per- $\alpha$  distillation ladder summary. 15/15 students evaluated with the canonical donor registry; “dev BLEU-4 (max)” is the training-time dev BLEU-4 recorded by the original training protocol (legacy validation-selected protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2; checkpoints were selected by training-time dev NLL, Sup. E).  $\alpha=0$  = hard-label only;  $\alpha=1$  = full soft-label KL. All  $\alpha<1.0$  yield negative gaps;  $\alpha=1.0$  produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). Train-pool readout uses the identical full-7,060, beam-3,  $[::2]$ -subsample protocol for every student and for the released evaluator (no 1,000-item-prefix mixing); the three  $\alpha=1.0$  students decode to empty or near-empty strings on both dev and test (uniform full-pool: 515/515 or 390/515 dev hypotheses empty, 641/641 or 483/641 test hypotheses empty) and score 0.00/0.0%; their dev BLEU-4 is degenerate and not reported.

| $\alpha$                      | $n$ | dev BLEU-4 (max)  | gap (range)      | train readout (BLEU / EM%) | reaches 10.5? |
|-------------------------------|-----|-------------------|------------------|----------------------------|---------------|
| 0.0 (hard-label)              | 3   | 10.51             | $[-1.71, -0.80]$ | 8.45–8.87 / 1.9%           | 1/3           |
| 0.25                          | 3   | 10.75             | $[-2.01, -0.77]$ | 7.81–8.63 / 1.5–1.9%       | 2/3           |
| 0.5                           | 3   | 10.41             | $[-1.38, -0.77]$ | 7.73–8.20 / 1.7–1.9%       | 0/3           |
| 0.75                          | 3   | 10.93             | $[-1.21, -0.81]$ | 6.59–7.01 / 1.3–1.4%       | 2/3           |
| 1.0 (full KL)                 | 3   | <i>degenerate</i> | $[0.00, 0.00]$   | 0.00 / 0.0% (empty)        | —             |
| Released (full 7,060, beam=3) | 1   | 13.38             | +10.24           | 78.82 / 70.7%              | —             |

**Table S37:** Canonical checkpoint registry (summary). 76 training runs across 12 families; 69 non-degenerate checkpoints enter the non-reproduction count (all 69 with strictly negative gaps; all share the clip=1.0 inference, resolved in Sup. U), 74 have a canonical PHX-public PURE-REC gap, and the released evaluator is the audit target. Auto-generated from the disk-scanned `results/canonical_checkpoint_registry.json`; full per-checkpoint details in Sup. D.

| Family                                                               | Runs | Trained | Has gap | Has dev | Enters                               |
|----------------------------------------------------------------------|------|---------|---------|---------|--------------------------------------|
| Reconstructions (paper-derived, 14)                                  | 14   | 14      | 14      | 14      | multiseed, bootstrap, PI, dose-resp. |
| Partially-corrected (config-faithful + step-corrected, 14)           | 14   | 14      | 14      | 14      | recipe-faithfulness evidence         |
| Joint-loss (greedy + beam-3, 16)                                     | 16   | 16      | 16      | 16      | protocol-robustness                  |
| Other diagnostic (confirmation + rescue + ladder + long-schedule, 8) | 8    | 8       | 6       | 8       | dose-resp.                           |
| Distillation (15)                                                    | 15   | 15      | 15      | 15      | ladder, dose-resp.                   |
| Joint-loss supplementary (1809/1810, 1)                              | 1    | 1       | 0       | 1       | (exploratory, not decoded)           |
| Total                                                                | 68   | 68      | 65      | 68      |                                      |
| Released original                                                    | —    | —       | 1       | 1       | audit target                         |

**Table S38:** Experimental protocol summary (moved from the main text).

| Setting          | Recognizer               | Layout    | Split size       | $N$ eval | $\alpha$      |
|------------------|--------------------------|-----------|------------------|----------|---------------|
| PHX-public       | SLRTP2025 evaluator      | 178-joint | 641 seq          | 641      | 0.88 (tuned)  |
| PHX-holdout-MSKA | mska_slr (phoenix-2014t) | 133-joint | 1,538 win        | 1,538    | 0.50 (frozen) |
| CSL-Daily        | mska (csl-daily)         | 133-joint | 256 of 1,176 win | 256      | 0.50 (frozen) |

**Table S39:** Competence rescue sweep: 20 runs across six hyperparameter families all fail the competence gate. Best variant: weight-decay 0, seed 202 (gate threshold defined in the main text).

| Variant family                                      | dev BLEU-4 range | min $ \Delta\text{BLEU-4} $ vs Original |
|-----------------------------------------------------|------------------|-----------------------------------------|
| lr $5 \times 10^{-4}$ / $2 \times 10^{-3}$ (8 runs) | 7.9–9.7          | 3.7                                     |
| batch size 128 / 512                                | 6.8–9.2          | 4.2                                     |
| dropout 0.2                                         | 6.9–7.5          | 5.9                                     |
| weight decay 0                                      | <b>9.8–9.9</b>   | <b>3.5</b>                              |
| label smoothing 0.1                                 | 6.8–7.4          | 6.0                                     |
| long schedule (600 epochs, lr $5 \times 10^{-4}$ )  | 8.2–8.3          | 5.1                                     |

**Table S40:** Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local REC). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

| Evaluator     | SEEN-PURE | UNSEEN-PURE | SEEN-REC       | UNSEEN-REC     | Estimand       |
|---------------|-----------|-------------|----------------|----------------|----------------|
| Original      | 23.02     | 8.51        | +10.24         | -4.27          | <b>+14.51</b>  |
| Seeds (range) | 6.83–9.09 | 6.59–7.98   | -1.85 to -0.80 | -1.94 to -0.27 | -0.15 to +1.22 |

**Table S41:** Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2026) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

| Reference condition                         | <i>N</i>                                                                     | REC       | PURE      | Gap                    |
|---------------------------------------------|------------------------------------------------------------------------------|-----------|-----------|------------------------|
| <i>Primary: matched confidence=1 subset</i> |                                                                              |           |           |                        |
| Original (speech-transcribed)               | 461                                                                          | 14.94     | 24.98     | +10.03                 |
| Human back-translation (conf=1)             | 461                                                                          | 7.35      | 8.30      | +0.95                  |
| Paired reference-frame contrast             | 9.08 BLEU pts (90.5% gap reduction), CI [7.44, 10.86]; ratio CI [0.81, 1.01] |           |           |                        |
| <i>Sensitivity: full 641 items</i>          |                                                                              |           |           |                        |
| Original (speech-transcribed)               | 641                                                                          | 12.78     | 23.02     | +10.24 [+8.88, +11.62] |
| Human back-translation (full)               | 641                                                                          | 5.66      | 6.52      | +0.86 [+0.38, +1.91]   |
| Paired contrast (sensitivity)               | 9.38 BLEU pts (91.6%)                                                        |           |           |                        |
| Human refs, reconstructions (6 seeds)       | 641                                                                          | 3.49–4.62 | 3.35–4.35 | −0.15 to −0.65         |

**Table S42:** Cross-evaluator re-decoding of the noisy-gloss retrieval outputs (641 PHX-public keys; §4.5). Gap = noisy-gloss – REC, corpus sacreBLEU-4 (flat-reference form, §3 protocol). The realistic retrieval pipeline is roughly even with REC under the released evaluator (-0.13), and every non-degenerate evaluator gives a negative gap; the five degenerate checkpoints (three  $\alpha=1.0$  distillation students, two smallest ladder fractions) decode empty or near-empty hypotheses and are excluded.

| Evaluator                              | <i>n</i> | Gap range                   | Mean gap |
|----------------------------------------|----------|-----------------------------|----------|
| Released                               | 1        | -0.13 (NG 12.65, REC 12.78) | -0.13    |
| Reconstructions                        | 14       | [-1.58, -0.37]              | -1.14    |
| Validation-freq-misread                | 4        | [-1.79, -1.00]              | -1.51    |
| Step-corrected (re-trained)            | 10       | [-1.36, -0.42]              | -0.85    |
| Joint-loss (greedy)                    | 8        | [-1.99, -0.84]              | -1.43    |
| Joint-loss (beam-3)                    | 8        | [-2.17, -0.48]              | -1.64    |
| Confirmation                           | 2        | [-0.95, -0.67]              | -0.81    |
| Long-schedule                          | 1        | [-1.79]                     | -1.79    |
| Distillation students ( $\alpha<1.0$ ) | 12       | [-1.94, -0.60]              | -1.23    |
| Rescue wd0                             | 1        | -1.49                       | -1.49    |
| Ladder (non-degenerate fractions)      | 2        | [-0.58, -0.07]              | -0.33    |

**Table S43:** Cross-evaluator re-decoding of the rebuilt USTC-MoE fixed outputs (497 of 641 PHX-public keys; §4.5). Gap = USTC-MoE – REC, corpus sacreBLEU-4. The rebuild decodes far below the officially reported leaderboard score (2.42 vs. 12.06), so it is pipeline-faithful but not score-faithful to the archived submission; every non-degenerate evaluator gives a negative gap. The five degenerate checkpoints (three  $\alpha=1.0$  distillation students, two smallest ladder fractions) decode empty or near-empty hypotheses and are excluded.

| Evaluator                              | $n$ | Gap range                        | Mean gap |
|----------------------------------------|-----|----------------------------------|----------|
| Released                               | 1   | -9.94 (USTC-MoE 2.42, REC 12.36) | -9.94    |
| Reconstructions                        | 14  | [-8.30, -5.91]                   | -7.45    |
| Validation-freq-misread                | 4   | [-7.77, -6.72]                   | -7.27    |
| Step-corrected (re-trained)            | 10  | [-8.29, -7.01]                   | -7.67    |
| Joint-loss (greedy)                    | 8   | [-9.52, -6.57]                   | -8.40    |
| Joint-loss (beam-3)                    | 8   | [-9.47, -7.98]                   | -8.91    |
| Confirmation                           | 2   | [-6.91, -6.65]                   | -6.78    |
| Long-schedule                          | 1   | [-7.61]                          | -7.61    |
| Distillation students ( $\alpha<1.0$ ) | 12  | [-8.26, -6.13]                   | -7.16    |
| Rescue wd0                             | 1   | -7.58                            | -7.58    |
| Ladder (non-degenerate fractions)      | 2   | [-5.47, -4.28]                   | -4.88    |

**Table S44:** Four-layer reproduction of the headline results from the artifact (§3). Layer 1 regenerates every paper number from committed per-item decodes; Layer 2 re-executes analyses from per-checkpoint data; Layer 3 re-decodes fixed outputs from checkpoint weights; Layer 4 retrains from scratch. **make core-audit** executes L1 only and is not an end-to-end reproduction from training data.

| Layer                       | Command                                                                                                                                            | Scope                                                                                                                                                          | Does NOT do                                                 |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| L1: number recomputation    | <code>make core-audit</code>                                                                                                                       | Regenerates canonical donor registry and all headline numbers from committed per-item decodes (asserts REC 12.78 / PURE 23.02 / gap +10.24 / CI [8.88, 11.62]) | Does not re-decode or retrain                               |
| L2: analysis re-execution   | <code>make round2-all;</code><br><code>make individual make targets</code><br>(donor-bootstrap, ref-frame-figure, experiment-C/A1/D, full-readout) | Re-runs every analysis from committed per-checkpoint data and hypotheses                                                                                       | Does not re-decode                                          |
| L3: model re-evaluation     | <code>scripts/decode_cells.py;</code><br><code>scripts/e_full_readout.py</code>                                                                    | Re-decodes fixed outputs from checkpoint weights                                                                                                               | Does not retrain; requires checkpoint weights + data bundle |
| L4: from-scratch retraining | <code>scripts/train_reconstruction.py;</code><br><code>launch_reconstructions.sh;</code><br>rescue/distillation/validation-freq-misread launchers  | Retrains and re-evaluates from the public recipe                                                                                                               | Requires PHOENIX-2014T licence, exact env, ~200 GPU-hours   |

**Table S45:** Per-checkpoint teacher-forced and free-decode diagnostic statistics. Teacher-forced token accuracy is the fraction of non-pad tokens where arg max of the teacher-forced logits matches the target; free-decode BLEU uses beam-3 autoregressive decoding on the full training pool (7,060) and full dev set (515) under the uniform protocol. Full data in `results/checkpoint_stats.json`.

| Checkpoint  | Train tok. acc. | Train NLL/tok | Train free-decode BLEU | Dev NLL/tok | Dev free-decode BLEU |
|-------------|-----------------|---------------|------------------------|-------------|----------------------|
| Released    | 0.965           | 0.158         | 78.82                  | 3.236       | 13.38                |
| Seed 101    | 0.570           | 1.822         | 8.54                   | 2.648       | 8.51                 |
| Seed 202    | 0.550           | 1.887         | 7.62                   | 2.675       | 6.88                 |
| Seed 303    | 0.596           | 1.723         | 8.24                   | 2.672       | 7.72                 |
| Seed 404    | 0.550           | 1.922         | 8.06                   | 2.645       | 7.33                 |
| Seed 505    | 0.564           | 1.870         | 8.58                   | 2.655       | 9.04                 |
| Seed 606    | 0.574           | 1.816         | 7.20                   | 2.672       | 6.49                 |
| Seed 707    | 0.590           | 1.718         | 9.25                   | 2.644       | 9.37                 |
| Seed 808    | 0.574           | 1.806         | 9.11                   | 2.654       | 8.97                 |
| Seed 909    | 0.565           | 1.851         | 8.58                   | 2.654       | 7.84                 |
| Seed 1001   | 0.584           | 1.767         | 8.63                   | 2.682       | 9.33                 |
| Seed 1102   | 0.584           | 1.759         | 8.94                   | 2.632       | 8.67                 |
| Seed 1203   | 0.574           | 1.793         | 8.38                   | 2.659       | 7.45                 |
| Seed 1304   | 0.564           | 1.852         | 8.47                   | 2.642       | 8.93                 |
| Seed 1405   | 0.564           | 1.826         | 7.34                   | 2.677       | 7.22                 |
| Reco. mean  | 0.572           | 1.815         | 8.35                   | 2.659       | 8.13                 |
| Reco. range | 0.550–0.596     | 1.718–1.922   | 7.20–9.25              | 2.632–2.682 | 6.49–9.37            |

**Table S46:** Complete retrieval  $\times$  construction design matrix explored on the released evaluator. Scores are corpus-level sacreBLEU-4 (0–100) on the canonical 641-item PHX-public set (REC=12.78). Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable (<sup>a</sup>), and 3 were not explored (\*, <sup>†</sup>); full reasons in `results/probe_multiplicity.json`. The headline probe (bold) was selected as the maximal-gap explored cell; its CI is post-selection descriptive. Two random-donor baselines (below the midrule) lie outside the  $4 \times 3$  matrix.

| Retriever                                                                                         | Construction           | Explored | Score        | Gap vs. REC               |
|---------------------------------------------------------------------------------------------------|------------------------|----------|--------------|---------------------------|
| <b>Source-text Jaccard</b>                                                                        | <b>Pure donor copy</b> | ✓        | <b>23.02</b> | <b>+10.24</b>             |
| Source-text Jaccard                                                                               | PT-composed            | ✓        | 18.86        | +6.08                     |
| Source-text Jaccard                                                                               | Oracle-gloss-comp.     | —        | —            | N/A <sup>a</sup>          |
| LaBSE                                                                                             | Pure donor copy        | ✓        | 19.2         | +6.42                     |
| LaBSE                                                                                             | PT-composed            | —        | —            | not explored*             |
| LaBSE                                                                                             | Oracle-gloss-comp.     | —        | —            | N/A <sup>a</sup>          |
| multi-SBERT                                                                                       | Pure donor copy        | ✓        | 16.5         | +3.72                     |
| multi-SBERT                                                                                       | PT-composed            | —        | —            | not explored*             |
| multi-SBERT                                                                                       | Oracle-gloss-comp.     | —        | —            | N/A <sup>a</sup>          |
| Oracle gloss                                                                                      | Pure donor copy        | ✓        | 11.4         | −1.38                     |
| Oracle gloss                                                                                      | PT-composed            | ✓        | 11.8         | −0.98                     |
| Oracle gloss                                                                                      | Oracle-gloss-comp.     | —        | —            | not explored <sup>†</sup> |
| <i>Reference-free retrieval and random baselines (outside the <math>4 \times 3</math> matrix)</i> |                        |          |              |                           |
| Pose-signature NN (no text)                                                                       | Pure donor copy        | ✓        | 0.45         | −12.33                    |
| Random donor                                                                                      | Pure donor copy        | ✓        | 0.90         | −11.88                    |
| Random donor                                                                                      | PT-composed            | ✓        | 0.77         | −12.01                    |

<sup>a</sup>Oracle-gloss-composed is definitionally tied to the oracle-gloss retriever, so it cannot be applied to a text-based retriever without changing the retriever. \*Not explored; deemed low priority after Jaccard pure-copy dominated. <sup>†</sup>Subsumed by Oracle-gloss  $\times$  PT-composed. Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable, and 3 were not explored. The headline probe was selected as the maximal-gap cell, so its CI is post-selection descriptive.

**Table S47:** Retrieval-query  $\times$  scoring-frame grid under the released evaluator (461-item subset; sacreBLEU-4; descriptive paired bootstrap CIs, 10,000 resamples). The response is positive only on the diagonal where the retrieval query coincides with the scoring reference. The two selectors pick the same donor for only 6.5% of queries; human-query donors still average Jaccard 0.30 to the original reference texts (canonical selector: 0.55; human-query selector against the human text: 0.33).

|                         | REC (recorded poses) | PURE, query=original | PURE, query=human    |
|-------------------------|----------------------|----------------------|----------------------|
| Score vs. original refs | 14.94                | 24.98                | 10.92                |
| Gap vs. REC             | —                    | +10.03 [8.41, 11.69] | −4.02 [−5.50, −2.62] |
| Score vs. human refs    | 7.35                 | 8.30                 | 11.64                |
| Gap vs. REC             | —                    | +0.95 [−0.14, 2.02]  | +4.29 [3.09, 5.49]   |

**Table S48:** Sensitivity grid (seed 42; protocols as in the main text). The CTC block varies the CTC denominator at clip 1.0; the clipping block varies `max_norm` with the per-sentence CTC default; the released row is the shipped evaluator. Readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate.

| Knob                       | Setting        | Dev BLEU-4 | Test REC | PURE  | Gap    | Readout (EM%) |
|----------------------------|----------------|------------|----------|-------|--------|---------------|
| CTC denom. (clip 1.0)      | sentences      | 12.05      | 13.11    | 10.32 | −2.80  | 11.96 (2.4%)  |
|                            | gloss tokens   | 10.79      | 10.42    | 9.01  | −1.40  | 10.06 (2.0%)  |
|                            | input frames   | 9.55       | 9.62     | 8.75  | −0.88  | 9.18 (1.9%)   |
|                            | raw sum        | 0.90       | 0.95     | 0.95  | +0.01  | 0.67 (0.0%)   |
| Clip $\infty$ ? (CTC sent) | 1.0 (faithful) | 12.05      | 13.11    | 10.32 | −2.80  | 11.96 (2.4%)  |
|                            | 2.0            | 17.17      | 16.35    | 15.90 | −0.45  | 23.65 (9.0%)  |
|                            | 3.0            | 17.13      | 16.78    | 21.82 | +5.04  | 53.21 (31.8%) |
|                            | 5.0            | 16.54      | 15.92    | 23.97 | +8.05  | 82.69 (68.4%) |
|                            | none (seed 42) | 16.32      | 15.08    | 26.27 | +11.19 | 99.98 (99.9%) |
|                            | none (seed 43) | 15.78      | 14.32    | 25.63 | +11.31 | 95.25 (90.5%) |
|                            | none (seed 44) | 15.74      | 15.96    | 26.27 | +10.31 | 99.98 (99.9%) |
| signjoey-exact             | none + raw sum | 16.04      | 15.12    | 25.80 | +10.68 | 97.60 (95.1%) |
| Released evaluator         |                | 13.38      | 12.78    | 23.02 | +10.24 | 78.8 (70.7%)  |